# MILITARY SPECIFICATION

## RADIO SET AN/GRC-103(V)4

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

### 1. SCOPE

1.1 This specification covers the following units of a radio relay equipment in the frequency range 1350 to 1850 MHz and designated Radio Set AN/GRC-103(V)4.

Receiver, Radio R-1329( )(P)/GRC-103(V)
Amplifier-Converter AM-4319( )/GRC-103(V)
Transmitter, Radio T-983( )(P)/GRC-103(V)
Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V)
Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)
Antenna AS-3047( )/GRC-103(V)
Mast AB-952( )/GRC-103(V)
Cable Assembly, RF CG-3444( )/U (1 ft. 6 in.)
Cable Assembly, RF CG-3444( )/U (3 ft. 6 in.)
Cable Assembly, RF CG-3443( )/U (80 ft.)
Adapter, Connector UG-1375( )/U
Cable Assembly, Special Purpose, Electrical, Branched CX-10763( )/GRC-103(V) (3 ft. 6 in.)

1.2 The radio set uses the following special accessories for extension of capabilities.

Extension Kit, Mast MK-1009( )/GRC-103(V)
Converter, Frequency, Electronic CV-2500( )/GRC
Cable Assembly, Special Purpose, Electrical CX-10879( )/GRC-103(V)
Dummy Load, Electrical DA-437( )/GRC-103(V)
Cable Assembly, Special Purpose, Electrical CX-10762( )/U (5 ft.)



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## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The follcwing documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-S-901 | Shock Tests- H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirement for (Navy) |
| MIL-P-11268 | Parts, Materials and Processes Used in Electronic Communication Equipment |
| MIL-M-13231 | Marking of Electronic Items |
| MIL-F-14072 | Finishes for Ground Electronic Equipment |
| MIL-E-15090 | Enamel, Equipment, Light-Gray (Formula No. 111) |
| MIL-R-45282 | Radio Set AN/GRC-103(V) (Units of); Packaging of |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |



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Military

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| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-252 | Wired Equipment, Classification of Visual and Mechanical Defects |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-781 | Reliability Tests: Exponential Distribution |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-1472 | Human Engineering Design Criteria for Military Systems, Equipment and Facilities |
| MIL-STD-1562 | Lists of Standard Microcircuits |

DRAWINGS

Electronics Command

| | |
|---|---|
| SC-A-46439 | List of Accessories for Package Tester |
| SC-DL-424153 | Adapter, Connector UG-1375/U |
| DL-SM-B-696006 | Receiver, Radio R-1329( )(P)/GRC-103(V) |
| SM-A-696070 | Test Procedure, Mast AB-952( )/GRC-103(V) |
| DL-SM-B-696151 | Mast AB-952( )/GRC-103(V) |
| DL-SM-B-696171 | Transmitter, Radio T-983( )(P)/GRC-103(V) |
| DL-SM-B-696175 | Receiver-Transmitter, Order Wire RT-773( )/GRC-103 |
| DL-SM-B-696203 | Cable Assembly, RF CG-3443( )/U |



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shall be observed, except when military operations prevent, when such railways

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| DL-SM-B-696207 | Cable Assembly, R.F. CG-3444( )/U |
| DL-SM-B-696210 | Cable Assembly, S.P. Electrical CX-10763( )/GRC-103(V) |
| SM-A-696443 | Test Specification-Amplifier-Frequency Multiplier |
| SM-A-698002 | Test Specification-Intermediate Frequency Amplifier |
| SM-A-698031 | Test Specification-Electrical Frequency Limiter-Discriminator |
| SM-A-698035 | Test Specification-Video Amplifier |
| SM-A-698241 | Test Specification-Pulse Form Restorer |
| SM-A-698317 | Test Specification-Monitor Panel |
| SM-A-698363 | Test Specification-Power Supply |
| SM-A-698372 | Test Specification-Power Supply |
| SM-A-698610 | Test Specification-Amplifier-Frequency Multiplier |
| SM-A-698644 | Alignment Procedure-Intermediate Frequency Amplifier |
| SM-A-698664 | Test Specification-Monitor Amplifier |
| SM-A-698685 | Test Specification-Distribution Box |
| SM-A-698695 | Test Specification-Monitor Panel |
| SM-A-698759 | Test Specification-Distribution Panel |
| SM-A-698776 | Test Specification-Alarm Control |
| SM-A-698789 | Test Specification-Distribution Box |
| SM-A-698790 | Test Specification-Monitor Panel |
| SM-A-698805 | Test Specification-Telephone Signal Converter |
| SM-A-698806 | Test Specification-Amplifier Assembly |
| SM-A-698807 | Test Specification-Rcvr-Tmtr Order Wire RT-773( )/GRC-103(V) |
| SM-A-698826 | Test Specification-Radio Receiver R-1329( )(P)/GRC-103(V) |
| SM-A-698831 | Test Specification-Radio Transmitter T-983( )(P)/GRC-103(V) |
| SM-A-710010 | Alignment Procedure-Voltage Regulator Sub-Assembly |
| SM-E-698042 | Random PCM Generator |



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| DL-SM-B-794068 | Amplifier-Converter AM-4319( )/GRC-103(V) |
| DL-SM-B-794070 | Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V) |
| DL-SM-B-794072 | Antenna AS-3047( )/GRC-103(V) |
| SM-A-794072 | Antenna AS-3047( )/GRC-103(V) |
| SM-A-794127 | Alignment Procedure-Amplifier-Converter AM-4319( )/GRC-103(V) |
| SM-A-794146 | Test Specification-Radio Set AN/GRC-103(V)4 |
| SM-A-794147 | Test Specification-Radio Set AN/GRC-103(V)4 (Receiver Section) |
| SM-A-794148 | Test Specification-Radio Set AN/GRC-103(V)4 (Transmitter Section) |
| SM-A-794152 | Alignment Procedure-Bandpass Filter |
| SM-A-794156 | Alignment Procedure-Amp-Freqy Multiplier AM-4323( )/GRC-103(V) |
| SM-A-794195 | Alignment Procedure-Bandpass Filter |
| SM-A-794205 | Alignment Procedure-Radio Frequency Amplifier |
| SM-A-794274 | Alignment Procedure-Radio Frequency Amplifier |
| SM-A-794289 | Test Procedure-Electronic Frequency Converter |
| SM-A-794315 | Alignment Procedure-Frequency Multiplier |
| SM-A-794356 | Alignment Procedure-Amplifier-Frequency Multiplier |
| SM-A-794410 | Test Procedure-Control-Indicator |
| SM-A-865041 | Test Specification-Electrical Frequency Synthesizer |

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and the drawing and data list as indicated, including the requirements for parts, materials and processes thereon.

| | |
|---|---|
| DL-SM-B-696006 | Receiver, Radio R-1329( )(P)/GRC-103(V) |
| DL-SM-B-794068 | Amplifier-Converter AM-4319( )/GRC-103(V) |
| DL-SM-B-696171 | Transmitter, Radio T-983( )(P)/GRC-103(V) |
| DL-SM-B-794070 | Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V) |
| DL-SM-B-696175 | Recv-Trans, Order Wire RT-773( )/GRC-103(V) |
| DL-SM-B-794072 | Antenna AS-3047( )/GRC-103(V) |
| DL-SM-B-696151 | Mast AB-952( )/GRC-103(V) |
| DL-SM-B-696207 | Cable Assy., R.F. CG-3444( )/U |
| DL-SM-B-696203 | Cable Assy., R.F. CG-3443( )/U |
| DL-SM-B-696210 | Cable Assy., S.P. Elec., CX-10763( )/GRC-103(V) |
| SC-DL-434153 | Adapter, Connector UG-1375( )/U |

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. Microcircuits shall be selected from MIL-STD-1562.



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3.4 Electrical characteristics. Radio Set AN/GRC-103(V)4 shall provide for multichannel, line-of-sight, two-way communication in the frequency range of 1350 - 1850 MHz (Channels 2300 - 3299). It is intended primarily for use as a radio link in a communication network which may include carrier telephone and teletypewriter sets. The radio set is used in pairs: one at each end of the line-of-sight path. The radio set may be used with 6/12 or 24-channel pulse code modulation (PCM) equipment or with 4-channel frequency division modulation (FDM) when associated with appropriate interface equipment. The system shall provide channel assignments on each half and full megahertz points (1/2 MHz spacing).

3.4.1 Transmitter section.

3.4.1.1 Output power. The normal transmitter RF output power, without the RF duplexer, shall be not less than 15 watts at any frequency in the range of 1350 to 1850 MHz. (See 4.16)

3.4.1.2 Low power alarm. A visual and audible alarm shall be provided in the transmitter. The visual alarm shall indicate any change in RF output power that is greater than 8 dB ±2 dB below the nominal power output rating of the transmitter. The switch associated with the audible alarm shall be such that: In one switch position, the alarm will sound when the visual alarms are lighted as in the case of a failure; in the second switch position, the alarm will sound when the light is not lighted. The audible alarm thus shall indicate a change in state of the visual signal.

3.4.1.3 Power dissipation. The transmitter output power shall not decrease below specification limits after:

a. Running the transmitter for 15 minutes with its output terminals short-circuited.



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b. Running the transmitter for 15 minutes with its output terminals open-circuited.

c. Running the transmitter for 30 minutes without drive.

3.4.1.4 Frequency setability. It shall be possible to set the transmitter to any desired frequency at 0.5 MHz intervals in the frequency range of 1350 to 1850 MHz with an accuracy of better than 0.002% corresponding to 20 Hz per MHz.

3.4.1.5 Transmitter frequency response. The overall frequency response of the transmitter when a modulation signal to produce 100 kHz peak deviation is applied to the video input shall conform to Table I.

TABLE I - TRANSMITTER FREQUENCY RESPONSE

| | Relative Attenuation (dB) Ref 10 kHz | | |
|---|---|---|---|
| Frequency kHz | Optimum | Minimum | Maximum |
| .125 | 1.0 | 0 | 1.5 |
| .200 | .5 | 0 | 1.0 |
| .250 | .2 | 0 | .4 |
| 1.0 | 0 | -0.2 | 0.2 |
| 10 | 0 (REF) | -- | -- |
| 30 | 0 | -0.2 | .2 |
| 60 | .1 | 0 | .3 |
| 120 | .15 | 0 | .5 |
| 240 | .7 | .5 | 1.5 |
| 360 | 1.5 | 0.6 | 2.3 |
| 480 | 2.7 | 1.9 | 4.2 |
| 600 | 4.2 | 3.4 | 6.2 |
| 720 | 6.0 | 5.3 | 8.8 |
| 840 | 8.2 | 7.5 | 11.5 |
| 960 | 10.7 | 8.5 | 15.0 |



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3.4.1.6 Spurious outputs. When measured with duplexer, the power output of any harmonic shall be at least 80 dB below that of the fundamental and the power output of any non-harmonic extraneous emission of a band outside plus and minus 15 MHz of the carrier shall be at least 80 dB below that of the fundamental. All spurious outputs over the frequency range $f_c \pm 15$ MHz shall be suppressed at least 60 dB, except for the range occupied by the modulation, where the spurious outputs shall not exceed the level of the modulation envelope.

3.4.1.7 FDM modulation requirements. With a 1 kHz sinusoidal signal of 191 mV applied to the transmitter video input, and the input control adjusted to give a reading of 50% on the panel meter, the transmitter peak deviation shall be between 95 kHz and 97 kHz. This requirement shall be met with the meter switch in the FDM position.

3.4.2 Receiver section. The following requirements shall be met, when tested with the duplexer.

3.4.2.1 Frequency response shaping. The overall frequency response of the receiver when a frequency modulated signal of constant deviation (100 kHz peak) is applied to the receiver antenna terminal and measured at the video output terminal shall conform to Table II.



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TABLE II - RECEIVER FREQUENCY RESPONSE

| Frequency kHz | Relative Attenuation (dB) Ref 10 kHz Optimum | Minimum | Maximum |
|---|---|---|---|
| 0.125 | 1.0 | -0.2 | 2.5 |
| 0.200 | 0.5 | -0.2 | 1.0 |
| 0.250 | 0.2 | -0.2 | 0.4 |
| 1 | 0 | -0.1 | 0.1 |
| 10 | 0 (REF) | -- | -- |
| 30 | 0 | -0.1 | 0.1 |
| 60 | 0.15 | 0.05 | 0.25 |
| 120 | 0.6 | 0.2 | 0.9 |
| 180 | 1.3 | 0.8 | 1.8 |
| 240 | 2.3 | 1.3 | 2.8 |
| 360 | 5.1 | 4.4 | 5.8 |
| 480 | 9.0 | 8.2 | 10.0 |
| 600 | 14.0 | 12.5 | 15.5 |
| 720 | 20.2 | 18.4 | 22.6 |
| 840 | 27.5 | 25.3 | 31.0 |
| 960 | 36.0 | 33.0 | 40.0 |

3.4.2.2 Noise figure. The noise figure of the receiver shall not exceed 8 dB over the operating frequency range. This value shall include the losses due to the receiver protection device and the duplexer. (See 4.15)

3.4.2.3 Demodulation. For received signals deviated +300 kHz, +75 kHz and +45 kHz, the video outputs shall be 0.265V +0.03V rms at the VIDEO connector across 50 ohms 0.189V ±0.02V rms at the FDM connector across 91 ohms and -10 dBm ±2 dB into 600 ohms at the OW connector, respectively.



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3.4.2.4 Received signal monitoring. A panel meter deflection of 60% or more shall be obtained at a received signal level of -50 dBm, and 10% or more at a level of -95 dBm.

3.4.2.5 IF bandwidth. The IF selectivity shall have a nominal 3 dB bandwidth of 750 kHz with the response down at least 60 dB at ±2.1 MHz or more from resonance.

3.4.2.6 Receiver protection. When the transmitter has been tuned for optimum output, it shall be possible to tune the receiver duplexer section through the transmitter frequency without damage to the receiver. The circuitry shall provide protection against external signals at any frequency with power level up to +45 dBm under all conditions of receiver tuning. Protection shall be provided against pulse modulated signals of any frequency of 1 KW peak power, duration 1 usec, duty cycle 0.001 and a rise time of 10 nsec from 10% to 90% amplitude.

3.4.2.7 Spurious response. The ratio of desired signal response to any spurious response shall be as follows:

a. For a maximum of 6 discrete responses, the ratio shall be at least 80 dB.

b. For all other responses, the ratio shall be at least 100 dB.

3.4.3 Terminal section (system requirements).

3.4.3.1 Input impedance. The impedance at the video input to the transmitter shall be 91 ohms ±10% resistive to at least 480 kHz and 91 ohms ±15% to at least 800 kHz.



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3.4.3.2 Output impedance. The output impedance of the receiver shall be 51 ohms ±10% resistive for the 24 channel PCM output to at least 800 kHz and for the regenerated 6/12 channel PCM output such that when terminated in a resistive load of 91 ohms ±5%, the waveform is as specified in 3.4.3.7.1.

NOTE: All received RF signal levels are measured at the antenna terminal of the duplexer.

3.4.3.3 Frequency response (PCM). The overall frequency response between transmitter video input and receiver video output relative to the response at 10 kHz shall conform to Table III.

TABLE III - PCM FREQUENCY RESPONSE

| | Relative Attenuation (dB) Ref 10 kHz | | |
|---|---|---|---|
| Frequency kHz | Nominal | Minimum | Maximum |
| .125 | 2.0 | 1.0 | 3.0 |
| .200 | 1.0 | 0 | 2.0 |
| .250 | .4 | 0 | .8 |
| .500 | 0 | -.1 | +.1 |
| 1.0 | 0 | -.1 | +.1 |
| 10 | 0 (REF) | -- | -- |
| 30 | 0 | -.1 | .2 |
| 60 | .2 | .1 | .4 |
| 120 | .7 | .5 | 1.0 |
| 180 | 1.6 | 1.3 | 2.2 |
| 240 | 2.9 | 2.5 | 4 |
| 360 | 6.5 | 5.7 | 8 |
| 480 | 11.7 | 10.2 | 13.5 |
| 600 | 18.2 | 16.0 | 21.0 |
| 720 | 26.2 | 23.1 | 30.5 |
| 840 | 35.7 | 31.7 | 41.0 |
| 960 | 46.7 | 41.2 | 52.5 |



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3.4.3.4 PCM modulation requirements. When the modulation sensitivity is adjusted to maximum, an input voltage of 1 volt peak-to-peak shall deviate the transmitter at least plus and minus 300 kHz peak. Also the metering circuitry shall enable the operator to adjust the peak deviation of the transmitter to plus and minus 180 kHz and plus and minus 300 kHz with an accuracy of plus 2.2 and minus 1.25 dB with inputs between 1 volt to 4 volts peak-to-peak. The transmitter shall be considered to have met this requirement if, when tested in accordance with the 12 and 24 CH PCM metering sections of the test specified in 4.19.3, it meets the specifications thereof.

3.4.3.5 Signal to noise (PCM). The transmitter shall be tuned to any of its operating frequencies and shall be operated at rated RF power output. A 10 kHz sinusoidal signal shall be applied at the video input terminals of the transmitter and the transmitter shall be adjusted for a peak deviation of plus and minus 180 kHz. The receiver output, when the receiver input is coupled to the transmitter through sufficient attenuation, shall be 0.45V +0.15V peak-to-peak at the unregenerated output terminals. Under these conditions, the ratio of average detected signal power to the average noise power at the unregenerated video output terminals of the receiver as measured with a flat weighted meter shall be at least 12 dB when the received RF signal is -94 dBm and at least 31 dB when the received RF signal is -74 dBm. With the transmitter adjusted for rated peak deviation (plus and minus 300 kHz), the receiver unregenerated video output shall be 3/4 ±1/4 volt peak-to-peak and the signal-to-noise ratio shall be at least 16 dB when the received RF signal is -94 dBm and at least 35 dB when the received signal is -74 dBm.

3.4.3.6 Distortion (PCM). The equipment shall be arranged as in 3.4.3.5 with the receiver coupled to the transmitter through sufficient attenuation so that the RF input at the receiver input terminals (including duplexer) is minus 54 dBm. A sinusoidal signal at any frequency within the range of 250 Hz to 100 kHz shall be applied at the video input terminals of the transmitter at a level of 2 volts peak-to-peak. The transmitter shall be adjusted for rated deviation and the receiver output shall be 3/4 plus or minus 1/4 volt peak-to-peak at the unregenerated video output terminals. Under these conditions, the total harmonic distortion measured at the unregenerated video output terminals shall not exceed a level 26 dB below the fundamental.



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3.4.3.7 Pulse regeneration (PCM). The receiver output circuits shall be capable of generating new pulses whose amplitude, timing and pulse shape are derived from the 6 channel or 12 channel signals briefly described below.

PCM CHARACTERISTICS

| | | | |
|---|---|---|---|
| Number of channels | 6 | 12 | 24 |
| Signal form | (full baud, unipolar binary) | | (Biternary combination of two 12 channel signals) |
| Bit rate | 288 | 576 | |
| Pulse width at half amplitude | 3.472 usec | 1.736 usec | |

3.4.3.7.1 Regenerated PCM. The regeneration shall eliminate all noise peaks less than 40% of the pulse amplitude for the range of unregenerated video output levels corresponding to input deviations of ±145 kHz peak minimum to ±230 kHz peak maximum. The regenerated signal amplitude shall be -0.1 volt ±0.1 volt for the '1' condition and -2.0 volts ±0.2 volt for the '0' condition, delivered into a load of 91 ohms ±5%, with no reversal of phase from the transmitter input or unregenerated video output. Pulse width shall be full interval with 10% to 90% rise time, 20 nsec minimum, 50 nsec maximum. The 50% point shall be 0 nsec minimum, 80 nsec maximum after the 50% point of the rising edge of the timing signal specified in 3.4.3.7.2.

3.4.3.7.2 Timing signal. Timing signals shall be extracted from the incoming pulse train and shall accompany the regenerated PCM to the next unit. The timing signal amplitude shall be 2 volts ±0.2V DC isolated, delivered into a load of 91 ohms ±5%. Pulse width at the 50% amplitude point shall be 80 nsec minimum, 150 nsec maximum. Rise time of the leading edge from the 10% point to the 90% point shall be 20 nsec minimum, 50 nsec maximum. Pulse rate shall be 576 kHz for both the 6 channel and 12 channel signals.



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3.4.3.7.3 Squelch and alarm. The squelch shall disable the regenerated PCM signal when the signal-to-noise ratio at the unregenerated video output is less than 7 dB ±2 dB when measured as in 3.4.3.5, with a reference deviation of ±180 kHz peak. The audible and visible alarm shall operate in conjunction with the squelch to indicate locally failure of the received signal from the remote station. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver, the audible alarm can be disabled. The sensitivity of the squelch circuit shall be such that the point of operation which indicates a received signal (lamp extinguished) shall be -98 dBm ±3 dB.

3.4.3.8 FDM demodulation requirements. With a signal deviated ±75 kHz applied to the receiver antenna terminal, the level measured across the FDM output connector terminated in 91 ohms shall be .189V ±.02V rms. The meter of the receiver shall be in the green area with the switch in the FDM position.

3.4.3.9 Duplexers. Insertion loss at the center frequency of the pass band of the transmitter section shall not exceed 2 dB over the operating frequency range. Insertion loss at the center frequency of the pass band of the receiver section shall not exceed 3 dB.

3.4.3.10 Duplexer discrimination. The discrimination of the filter for the transmitter section shall be a minimum of 18 dB at ±20 MHz and for the receiver section a minimum of 66 dB at ±20 MHz.

3.4.3.11 Voltage standing wave ratio. The VSWR measured at the antenna terminal of the duplexer, with the transmitter terminal terminated in a 50 ohm load and the receiver connected to the receiver connection, shall not exceed 2.2:1 at any transmitted frequency within the operating frequency range as specified in 3.4.



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3.4.3.12 Systems level stability. The transmitter and receiver shall be connected through an attenuator with sufficient attenuation such that the power input to the receiver input terminals is minus 60 dBm. With the transmitter modulated as described in 3.4.3.4 (PCM modulation requirements) and the overall system adjusted for nominal gain, the following conditions shall be met: The system gain shall not vary more than 1 dB in any eight hour period between the 91 ohm video input and 51 ohm video output terminals of the receiver when the received signal is varied between -95 dBm and -45 dBm under any combination of temperature, humidity or line voltage variation as specified herein.

3.4.4 Order wire.

3.4.4.1 Order wire operation. The order wire unit shall meet the following requirements to be compatible with Multiplexers TD-202( )/G, TD-204( )/G, and TD-754( )/G.

3.4.4.1.1 Tone generator. When the tone generator is activated, the level measured at the TRANS "A" & TRANS "B" terminals of the order wire unit shall be -10 dBm ±1 dB.

3.4.4.1.2 Signal receiver detector sensitivity. When a -10 dBm ±7 dB, 1600 Hz ±4 Hz signal is applied to either the REC A or REC B terminals of the order wire unit, the signal lamp shall light and the buzzer sound.

3.4.4.2 Receiver demodulation order wire. With an RF signal deviated ±45 kHz peak applied to the receiver antenna terminal, the level measured at the receiver order wire shall be -10 dBm ±2 dB.

3.4.4.3 Transmitter deviation order wire. A 1000 Hz sinusoidal signal at a level of zero dBm applied to the microphone input terminals of the order wire unit shall produce a peak deviation of the transmitter between 38 kHz and 59 kHz. A 1000 Hz signal of -10 dBm applied to the transmitter order wire connector shall produce a peak deviation of between 39 kHz and 58 kHz.



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3.4.4.4 Signal-to-noise (order wire). The transmitter and receiver shall be connected together in a systems arrangement with sufficient attenuation between them such that the power measured at the input terminals of the receiver is -24 dBm. With the transmitter modulated as in 3.4.4.3 the signal-to-noise ratio measured with FIA weighting at the local order wire terminals (receive phone terminals) shall be at least 29 dB at any operating frequency in the presence of a normal 6/12 channel PCM signal and at least 40 dB in the FDM mode of operation. When measured with C-message weighting the signal-to-noise ratio shall be at least 27.5 dB and 38.5 dB, respectively.

3.4.4.5 Peak limiting (order wire). The limiting action to be provided in the transmitter for the order wire shall be such as to compress test tone power (10 kHz) inserted at the microphone terminals in accordance with Table IV.

TABLE IV - ORDERWIRE PEAK LIMITING

| Limiting Characteristic, Insert Path Measured Between MIKE Terminal and TRAN 'A' or TRAN 'B' Terminals | |
|---|---|
| INPUT POWER | COMPRESSION |
| -10 dBm | None |
| -5 dBm | None |
| 0 dBm | Less than 0.5 dB |
| +5 dBm | 3.5 dB ±1 dB |
| +10 dBm | 7.5 dB ±2 dB |
| +15 dBm | 10.0 dB ±3 dB |

3.4.4.6 Frequency response (order wire system). The overall frequency response characteristic between the local microphone input and the phone output terminals of the order wire unit at a distant station shall be plus 1 to minus 3 dB from 300 to 1800 Hz, referred to the 1 kHz response at an RMS level of zero dBm, when the equipment is employed in a full duplex PCM system.



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3.4.4.7 Frequency response (order wire unit). The order wire unit RT-773( )/GRC-103 shall meet the following frequency response characteristics (Table V) when measured with input levels specified in 3.4.4.11.

TABLE V - OW FREQUENCY RESPONSE

| PATH | FROM | TO | FREQUENCY RANGE | RESPONSE REF: 1 kHz |
|---|---|---|---|---|
| 'THRU' | REC 'A' | TRAN'B' | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.0 dB |
| 'THRU' | REC 'B' | TRAN 'A' | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.0 dB |
| DROP | REC 'A' | PHONE terminal | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.5 dB |
| DROP | REC'B' | PHONE terminal | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.5 dB |
| INSERT | MIKE terminal | TRAN 'A' | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.0 dB |
| INSERT | MIKE terminal | TRAN 'B' | 200 Hz<br>-2.0 kHz | +0.5 dB<br>-1.0 dB |
| SIDETONE | MIKE terminal | PHONE terminal | 200 Hz<br>-2.0 kHz | ±1.0 dB |

3.4.4.8 Distortion (order wire system). With the equipment arranged as in 3.4.4.4, the total harmonic distortion measured at the order wire output terminals shall not exceed 5%.



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3.4.4.9 Distortion (order wire unit). The distortion of the order wire unit when measured with a 1000 Hz signal as specified in 3.4.4.7 shall be as follows:

| | | |
|---|---|---|
| 'THRU' PATHS | REC 'A' to TRAN 'B' | Less than 1% |
| | REC 'B' to TRAN 'A' | Less than 1% |
| INSERT PATHS | MIKE terminal to TRAN 'A' | Less than 2% |
| | MIKE terminal to TRAN 'B' | Less than 2% |
| DROP PATHS | REC 'A' to PHONE terminal | Less than 1% |
| | REC 'B' to PHONE terminal | Less than 1% |

3.4.4.10 Net loss (order wire system). The net loss between the local microphone input and the phone output at a distant station shall be 18 dB $\pm$3 dB.

3.4.4.11 Order wire unit levels. The order wire unit levels when measured with a 1000 Hz signal shall conform to Table VI:



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TABLE VI - OW UNIT LEVELS

| FROM | TO | OUTPUT LEVEL | INPUT LEVELS |
|---|---|---|---|
| REC 'A' | TRAN 'B' | -10 dBm ±0.1 dB (adjustable) | -10 dBm |
| REC 'B' | TRAN 'A' | -10 dBm ±0.1 dB (adjustable) | -10 dBm |
| REC 'A' | PHONE terminal | -18 dBm ±1.0 dB | -10 dBm |
| REC 'B' | PHONE terminal | -18 dBm ±1.0 dB | -10 dBm |
| MIKE terminal | TRAN 'A' | -10 dBm ±1.0 dB | 0 dBm |
| MIKE terminal | TRAN 'B' | -10 dBm ±1.0 dB | 0 dBm |
| MIKE terminal | PHONE terminal | -24 dBm ±2.0 dB (sidetone) | 0 dBm |
| REC 'A' | TRAN 'A' | -55 dBm +10.0 dB (crosstalk) | -10 dBm |
| REC 'B' | TRAN 'B' | -55 dBm ±10.0 dB (crosstalk) | -10 dBm |

3.4.5 Antenna.

3.4.5.1 Symmetry. The ratio of the half-power (-3 dB) E- and H-plane beamwidth shall not exceed 2.0:1 over the frequency range.

3.4.5.2 Squint. The gain of the antenna measured in the direction of the normal from the antenna shall not differ from the maximum gain measured at any frequency in the operating range by more than 0.5 dB.



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3.4.5.3 Wide angle radiation. Radiation at all angles greater than 25 degrees away from the axis of the main beam shall be at least 13 dB below the main beam.

3.4.5.4 Gain. The gain of the antenna relative to an isotropic radiator shall be at least 18 dB at 1350 MHz and 20 dB at 1850 MHz. At intermediate frequencies, the gain shall exceed the values linearly interpolated between the frequencies specified.

3.4.5.5 Voltage standing wave ratio. The VSWR measured at the antenna terminals shall not exceed 1.75:1 over the operating frequency range.

3.4.5.6 Impedance. The nominal input impedance of each antenna shall be 50.0 ohms.

3.5 Mast.

3.5.1 Erection. The mast shall be capable of erection with the antenna and transmission line installed in winds of twenty-five miles per hour with the launcher guyed and all other guy wires slack.

3.5.2 Operation. The mast assembly, erected with guy wires tensioned and antenna and transmission line installed, shall withstand the following conditions without permanent deformation and maintain the antenna azimuth within $\pm 5°$ of any initial setting.

a. Wind velocities up to 75 mph or;

b. Wind velocities up to 50 mph with a one-half inch radial ice load on the entire mast-antenna installation.



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3.6 Finish, protective. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. Unless otherwise specified, the final paint film on Type 1 surface (see MIL-F-14072), exposed to view during operation, shall be a smooth, semi-gloss (olive drab) green, in color conforming to No. 24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the Contracting Officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact. The top coat of paint on the front panel of the equipment shall be semi-gloss gray enamel in accordance with MIL-E-15090.

3.7 Marking.

3.7.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.7.2 Visibility. Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

3.7.3 Chassis marking. All chassis identification, parts identification, warning notes, etc., shall be applied by the silk screen process or processes giving the same degree of legibility, placement accuracy and permanence. The marking shall be in accordance with the drawings.



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3.7.4 Serial numbers. The following items shall have serial numbers and name plates.

Transmitter, Radio T-983( )(P)/GRC-103(V)

Amplifier, Frequency Multiplier AM-4323( )/GRC-103(V)

Receiver, Radio R-1329( )(P)/GRC-103(V)

Amplifier-Converter AM-4319( )/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-3047( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

3.7.4.1 Composition of serial numbers. The serial numbers used for this equipment shall be ten digits. The first five digits shall be the Federal identification number of the prime manufacturer. The sixth digit shall be a code letter to indicate the production contract with said manufacturer. A for first production, B for second production, etc. The final four digits shall be progressively from 0000 to 9999 as required. For substantially continuous production on more than one contract, consecutive numbering without change of letter designator may be used upon application to and approval by the contracting officer. Records shall be maintained and furnished the Government of the first and last serial number of each unit, assembly, subassembly, and module furnished on the contract.



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3.8 Power requirements. The equipment shall operate from 115 volt RMS, 47.5 Hertz to 420 Hertz, single phase source, having a voltage variation of plus or minus 10%. The total power drain for a transmitter and receiver shall not exceed 400 watts at normal operating conditions.

3.9 Mechanical interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically interchangeable without modification or adjustment except operator adjustments of such items or of the equipment. (See 4.21.1).

3.10 Electrical interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be functionally interchangeable without modification or adjustment except operator adjustments of such items or of the equipment. Units, assemblies, subassemblies and replaceable parts shall meet the electrical requirements of their respective drawings and the test of 4.19 and 4.21.2.

3.11 Service conditions.

3.11.1 Temperature.

3.11.1.1 Low. During testing as specified in 4.7.4.1, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.11.1.2 High. During testing as specified in 4.7.4.2, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.



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3.11.2 Moisture resistance (humidity). During testing as specified in 4.7.5, the equipment shall meet the performance requirements of 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1. Upon completion of tests there shall be no evidence of corrosion of degradation of finishes.

3.11.3 Elevation (altitude). The equipment shall meet full specification performance of 4.9.2 during and after testing as specified in 4.7.6.

3.11.4 Vibration, bounce and shock. The equipment shall be capable of meeting the requirements of Table VII.

TABLE VII - BOUNCE AND SHOCK

| Paragraph Heading | Test Para. | Performance after test |
|---|---|---|
| Bounce | 4.7.1 | Specified performance (Notes A & B) |
| Shock; ballistic | 4.7.2 | No physical damage. Operable (Note B). |
| Shock; bench-handling | 4.7.3 | Any physical damage shall be minor only. Specified performance (Note B). No physical damage. |

NOTE A: The amplitude of vibration of any part, sub-assembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.



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NOTE B: The equipment shall meet specified performance for the following:

a. Paragraph 4.2 of SM-A-794148 except at 115 Volts, 60 Hz.

b. Paragraph 4.5 of SM-A-794147.

c. Paragraph 4.1 of SM-A-794146.

d. Paragraph 4.4 of SM-A-794146.

e. Paragraph 4.10 of this specification.

3.12 Electromagnetic interference. The equipment shall meet the following emission and susceptibility requirements of MIL-STD-461, Notice 4:

| | | | |
|---|---|---|---|
| CE04 | CS01 | RE02 | RS03.2 |
| CE06 (key up only) | CS02 | RE02.1 | |
| | CS06 | | |

3.12.1 Bonds and grounds. Bonding for equipment current path return, RF potentials, and shock hazard shall be so installed that expansion, contraction or movement incident to normal service use will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance properties from the immediate area of contact. Bonding impedance shall not exceed 2.5 milliohms. Case mounted connectors shall have an approved RF conductive finish. Bonding impedance of the connector shell to the case shall not exceed 2.5 milliohms. Shielded cables shall be terminated with connectors having an approved RF conductive finish and an EMI backshell for 360° peripheral bonding of the shield.



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3.13 Preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the preconditioning of 4.6.

3.14 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Soldering and brazing
Cleaning of units.
General requirements for securing of parts.
Process of welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

3.15 Systems safety engineering.

3.15.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454 and the paragraph entitled "Hazards and Safety" of MIL-STD-1472. Compliance with these requirements will be verified through a visual inspection. (See 4.20)

3.15.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).



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3.16 Reliability.

3.16.1 Numerical reliability requirement. The equipment shall have a mean-time-between-failure (MTBF) of 1200 hours when operated in accordance with Appendix I to this specification as follows:

a. Temperature: 40° ±5°C

b. Temperature cycling: None

c. Vibration: 2.2G ±10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment. The duration of vibration shall be for two minutes of every operating hour during one eight-hour period each week.

d. Equipment on-off cycling: Equipment shall be operated continuously for periods of 144 hours and shall be switched off for periods of eight hours.

e. Input voltage: Nominal specified voltage ±5%.

f. Input voltage cycling: None

3.17 Operation prior to packaging. Each complete equipment shall meet a final operational test in accordance with 4.10 and approved test procedure after all other inspections and tests have been completed to determine that the unit is operational prior to packaging. (See 6.4)

3.18 Technical literature, tools, and running spare parts. Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include preparation for delivery.

b. Inspection covered by subsidiary documents. (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.24).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish seven (7) first article units of the complete Radio Set AN/GRC-103(V)4.



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4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections for Group A, Group B, Group C and Group D (see Table VIII, IX, X and XI respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units; and designated units for specific subgroups within Group C in accordance with 4.5.3.1 and Group D.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

| ITEM | WHERE REQUIRED |
|---|---|
| Construction | 3.2 |
| Parts, materials and processes | 3.3 |
| Finish | 3.6 |
| Marking | 3.7 |
| Workmanship | 3.14 |

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspections specified in 4.4, 4.6, 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.



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shall be applied in accordance with the requirements of the contract.

4.5.1 Group A inspection. Each unit (and each subassembly and assembly identified in Table VIII) on contract or purchase order shall be inspected for conformance to the inspections specified in Table VIII. The order of inspection of the subassemblies, assemblies and units (end equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the subassemblies and assemblies identified in Table VIII shall be subjected to a 100-percent visual/mechanical and electrical inspection by the contractor for conformance to the inspection requirements provided in Table VIII.

b. After completion of the above 100-percent inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100 percent inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105, using general inspection levels and AQL's indicated in Table II therein (quality assurance verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Subassemblies and assemblies from the discrete lots which have passed the above quality assurance verification inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 4.6 and a 100 percent visual/mechanical and electrical inspection in accordance with Table VIII in that order.

NOTE: Subassemblies and assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100 percent Group A inspection of the end equipment.



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d. After completion of the above 100 percent inspection by the contractor, discrete lots shall be formed from units which passed the 100 percent inspection and these shall be subjected to the contractor's sampling inspection utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table VIII. Group A electrical inspection shall be performed in the order shown in Table VIII with the exception that the operational test (4.10) shall be last. (Also see 6.6).

NOTE: Subassemblies and assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's sampling inspection of the end equipment.



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TABLE VIII - GROUP A INSPECTION

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Visual and Mechanical | | | | |
| Receiver, Radio (Unit 1) R-1329( )(P)/GRC-103(V) | 3.2,3.3, 3.6,3.7, 3.14 | 4.20 | 1% | 4% |
| Synthesizer (1RE1A2) | | | 1% | 4% |
| Panel, Monitor (1RE1A1) | | | 1% | 4% |
| Power Supply (1RE1PS1) | | | 1% | 4% |
| Restorer, Pulse Form (1RE1A3) | | | 1% | 4% |
| Limiter-Discriminator (1RE1A4) | | | 1% | 4% |
| Amplifier-Frequency Multiplier (1RE1A5) | | | 1% | 4% |
| Amplifier, Video (1RE1AR1) | | | 1% | 4% |
| Amplifier, Intermediate Frequency (1RE1AR2) | | | 1% | 4% |
| Case, Receiver (1A2) | | | 1% | 4% |
| Distribution Box (1A2A1) | | | 1% | 4% |
| Amplifier-Converter (Unit 39) AM-4319( )/GRC-103(V) | 3.2,3.3, 3.6,3.7, 3.14 | 4.20 | 1% | 4% |
| Electronic Component Assembly (39A1) | | | 1% | 4% |
| Converter, Frequency, Electronic (39A2) | | | 1% | 4% |
| Frequency Multiplier (39A3) | | | 1% | 4% |
| Control-Indicator (39A4) | | | 1% | 4% |
| Amplifier, Radio Frequency (39AR1) | | | 1% | 4% |
| Amplifier, Intermediate Frequency (39AR2) | | | 1% | 4% |
| Bandpass Filter (39FL1, FL2) | | | 1% | 4% |
| Bandpass Filter (39FL3) | | | 1% | 4% |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Transmitter, Radio (Unit 5) T-983( )(P)/GRC-103(V) | 3.2,3.3, 3.6,3.7 3.14 | 4.20 | 1% | 4% |
| Synthesizer (5TR1A2) | | | 1% | 4% |
| Panel, Monitor (5TR1A1) | | | 1% | 4% |
| Power Supply (5TR1PS1) | | | 1% | 4% |
| Control, Alarm (5TR1A3) | | | 1% | 4% |
| Amplifier-Frequency Multiplier (5TR1A4) | | | 1% | 4% |
| Amplifier-Monitor (5TR1A5) | | | 1% | 4% |
| Case,Transmitter (5A2) | | | 1% | 4% |
| Distribution Box (5A2A1) | | | | |
| Amplifier-Frequency Multiplier (Unit 40) AM-4323( )/GRC-103(V) | 3.2,3.3, 3.6,3.7, 3.14 | 4.20 | 1% | 4% |
| Electronic Component Assembly (40A1) | | | 1% | 4% |
| Amplifier-Frequency Multiplier (40A2) | | | 1% | 4% |
| Regulator,Voltage (40A3) | | | 1% | 4% |
| Control-Indicator (40A4) | | | 1% | 4% |
| Amplifier, Radio Frequency (40AR1) | | | 1% | 4% |
| Filter, Bandpass (40FL1) | | | | |
| Receiver-Transmitter, Order Wire (Unit 9) RT-773( )/GRC-103(V) | 3.2,3.3, 3.6,3.7, 3.14 | 4.20 | 1% | 4% |
| Panel,Distribution (9A1) | | | 1% | 4% |
| Panel,Monitor (9A2) | | | 1% | 4% |
| Converter, Telephone Signal (9A3) | | | 1% | 4% |
| Amplifier Assembly (9A4) | | | 1% | 4% |
| Case, Receiver-Transmitter Order Wire (9MP1) | | | | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Antenna (Unit 41) AS-3047( )/GRC-103(V) | 3.2,3.3, 3.6, 3.7, 3.14 | 4.20 | 1% | 4% |
| Antenna Feed (41A1) | | | 1% | 4% |
| Antenna Reflector (41A2) | | | 1% | 4% |
| Mast (Unit 12) AB-952( )/GRC-103(V) | 3.2,3.3, 3.6,3.7, 3.14 | 4.20 | 1% | 4% |
| Elevator, Antenna (12A1A1) | | | 1% | 4% |
| Mast Section (12A1) | | | 1% | 4% |
| Accessory Kit, Mast (12A2) | | | 1% | 4% |
| Bag, Mast Accessories | | | 1% | 4% |
| Guy | | | 1% | 4% |
| Ring, Guy Attachment | | | 1% | 4% |
| Spike, Common | | | 1% | 4% |
| Adapter, Antenna | | | 1% | 4% |
| Anchor, Guy | | | 1% | 4% |
| Hammer | | | 1% | 4% |
| Strap Wrench | | | 1% | 4% |
| Cable Assemblies | 3.2,3.3, 3.4,3.7, 3.14 | 4.20 | | |
| Cable Assy, R.F. CG-3443/U | | | 1% | 4% |
| Cable Assy, R.F. CG-3444/U | | | 1% | 4% |
| Cable Assy, S.P. Elec. CX-10763( )/GRC-103(V) | | | 1% | 4% |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Electrical (Subassemblies) | | | | |
| Receiver, Radio (Unit 1) R-1329( )(P)/GRC-103(V) | 3.10 | 4.19.1 | | |
| Dimmer Switch<br>Wideband Noise<br>Frequency Deviation<br>FDM Output<br>OW Output<br>PCM Regeneration<br>PCM Squelch<br>PCM Mute<br>Receiver Squelch<br>Receiver Signal Monitor<br>FDM Return Loss<br>Video Return Loss<br>Frequency Generation<br>Power Output<br>Video Frequency Response | | | 1% FOR THE GROUP | |
| Amplifier-Converter (Unit 39) AM-4319( )/GRC-103(V) | 3.10 | 4.19.2 | | |
| Receiver Setability<br>Input VSWR (Rcvr Section)<br>Input VSWR (Xmtr Duplexer)<br>High Signal Alarm<br>AGC Test<br>Noise Figure<br>Transmitter Duplexer Loss | | | 1% FOR THE GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Transmitter, Radio (Unit 5) T-983( )(P)/GRC-103(V) | 3.10 | 4.19.3 | | |
| Voltage Stability<br>Video Input and Deviation Metering<br>Power Output<br>Overheat Alarm<br>Low Power Alarm<br>Output Frequency Settability<br>Amplitude-Frequency Response<br>Video Input Match | | | 1% FOR THE GROUP | |
| Amplifier-Frequency Multiplier (Unit 40) AM-4323()/GRC-103(V) | 3.10 | 4.19.4 | | |
| Power Output and Tracking<br>Cathode and Heater Voltages<br>ALC Voltage | | | 1% FOR THE GROUP | |
| Receiver, Radio (1RE1) | | | | |
| Synthesizer (1RE1A2) | 3.10 | 4.19.5 | | |
| Frequency Check<br>Sync. Alarm<br>Power Output<br>Metering | | | 1% FOR THE GROUP | |
| Panel, Monitor (1RE1A1) | 3.10 | 4.19.8 | | |
| Continuity Check | | | 1% | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Power Supply (1RE1PS1) | 3.10 | 4.19.9 | | |
| Initial Calibration<br>Meter Check<br>Regulation<br>Load and Regulation<br>Unregulated Supply<br>Ripple Measurement (Unreg.)<br>Ripple Measurement (Reg.)<br>High Frequency Instability | | | 1% FOR THE GROUP | |
| Restorer, Pulse Form (1RE1A3) | 3.10 | 4.19.6 | | |
| Timing Pulse<br>PCM Regenerator<br>OW Noise Test<br>OW Distortion | | | 1% FOR THE GROUP | |
| Limiter-Discriminator (1RE1A4) | 3.10 | 4.19.7 | | |
| Demodulation Linearity<br>Deviation Sensitivity<br>Frequency Response | | | 1% FOR THE GROUP | |
| Amplifier-Frequency Multiplier (1RE1A5) | 3.10 | 4.19.10 | | |
| Harmonic Level | | | 1% | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Amplifier, Video (1RE1AR1) | 3.10 | 4.19.11 | | |
| Gain<br>24 Ch. PCM Adjust<br>Low Signal Alarm<br>Off Frequency Alarm<br>Distortion<br>Continuity Check | | | 1% FOR THE GROUP | |
| Amplifier, Intermediate Frequency (1RE1AR2) | 3.10 | 4.19.12 | | |
| Gain (without AGC)<br>Frequency Response<br>AGC Limited Output Level | | | 1% FOR THE GROUP | |
| Case, Receiver (1A2) | | | | |
| Distribution Box (1A2A1) | 3.10 | 4.19.13 | | |
| Continuity Check | | | 1% | |
| Amplifier-Converter (Unit 39) AM-4319( )/GRC-103(V) | | | | |
| Converter, Frequency, Electronic (39A2) | 3.10 | 4.19.14 | | |
| Conversion Gain<br>Image Rejection<br>L.O. Metering<br>L.O. Input VSWR<br>Noise Figure<br>R.F. Input VSWR<br>High Signal Alarm<br>I.F. Output VSWR | | | 1% FOR THE GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Frequency Multiplier (39A3) | 3.10 | 4.19.15 | | |
| Input VSWR<br>ALC Setting<br>Output Power and Input VSWR<br>Harmonic Rejection | | | 1% FOR THE GROUP | |
| Control-Indicator (39A4) | 3.10 | 4.19.16 | | |
| Coding Check | | | 1% | |
| Amplifier, Radio Frequency (39AR1) | 3.10 | 4.19.17 | | |
| Input VSWR<br>Output Power<br>Metering and ALC | | | 1% FOR THE GROUP | |
| Amplifier Intermediate Frequency (39AR2) | 3.10 | 4.19.18 | | |
| Frequency Response<br>Output VSWR<br>AGC<br>Noise Figure | | | 1% FOR THE GROUP | |
| Bandpass Filter (39FL1, 39FL2) | 3.10 | 4.19.19 | | |
| Q Bandwidth<br>Coupling Bandwidth<br>VSWR Bandwidth and Passband<br>Insertion Loss<br>Tracking<br>Discrimination | | | 1% FOR THE GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Bandpass Filter (39FL3) | 3.10 | 4.19.20 | | |
| Q Bandwidth<br>Coupling Bandwidth<br>VSWR and VSWR Bandwidth<br>Insertion Loss<br>Discrimination<br>Tracking | | | 1% FOR THE GROUP | |
| Transmitter, Radio (5TR1) | | | | |
| Panel, Monitor (5TR1A1) | 3.10 | 4.19.21 | | |
| Continuity Check | | | 1% | |
| Synthesizer (5TR1A2) | 3.10 | 4.19.5 | | |
| Frequency Check<br>Sync. Alarm<br>Power Output<br>Metering | | | 1% FOR THE GROUP | |
| Control, Alarm (5TR1A3) | 3.10 | 4.19.22 | | |
| Low Power Alarm and Backlash | | | 1% | |
| Amplifier-Frequency Multiplier (5TR1A4) | 3.10 | 4.19.23 | | |
| Frequency Response<br>Output Level<br>Power Output (With ALC)<br>Harmonic Suppression<br>Input VSWR | | | 1% FOR THE GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Amplifier-Monitor (5TR1A5) | 3.10 | 4.19.24 | | |
| Operational Check<br>Metering Check<br>Frequency Response | | | 1% FOR THE GROUP | |
| Power Supply (5TR1PS1) | 3.10 | 4.19.25 | | |
| Insulation and Resistance<br>Preliminary Check<br>Meter Reading<br>Input Rectification<br>High Tension<br>Rise Time and Frequency Measurement at 5kHz Converter<br>Rise Time and Frequency Measurement at 400 Hz Converter<br>26Vdc Unregulated Supply<br>Filament Supply<br>28Vdc/12Vdc Regulated Supply<br>High Frequency Instability<br>Constant Current<br>High Tension Metering Circuit Calibration<br>5kHz Transistor Saturation<br>Blower Voltage and Alarm Supply<br>Hum Measurement (120 Hz)<br>Spike Measurement (5 kHz)<br>Input AC Current<br>Short-Circuit Protection Check<br>Starting-Circuit Check | | | 1% FOR THE GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Case, Transmitter (5A2) | | | | |
| Distribution Box (5A2A1) | 3.10 | 4.19.26 | | |
| Continuity Check | | | 1% | |
| Amplifier-Frequency Multiplier (Unit 40) AM-4323( )/GRC-103(V) | | | | |
| Amplifier-Frequency Multiplier (40A2) | 3.10 | 4.19.27 | | |
| Tuning Law Alignment<br>Input VSWR Adjustment<br>Regulator Current Alignment<br>Output Power and Input VSWR Checks<br>Harmonics Rejection<br>Metering and ALC Control<br>Switching Signals Check | | | 1% FOR THE GROUP | |
| Regulator, Voltage (40A3) | 3.10 | 4.19.28 | | |
| Regulation<br>ALC<br>Cathode Current Control<br>Metering | | | 1% FOR THE GROUP | |
| Control-Indicator (40A4) | 3.10 | 4.19.16 | | |
| Coding Check | | | 1% | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Amplifier, Radio Frequency (40AR1) | 3.10 | 4.19.29 | | |
| Driver Stage Alignment<br>Output Stage Alignment<br>Output Power, Cathode Voltage, Tuning Law<br>Input VSWR | | | 1% FOR THE GROUP | |
| Bandpass Filter (40FL1) | 3.10 | 4.19.19 | | |
| Q Bandwidth<br>Coupling Bandwidth<br>VSWR Bandwidth and Passband<br>Insertion Loss<br>Tracking<br>Discrimination | | | 1% FOR THE GROUP | |
| Receiver-Transmitter, Order Wire (Unit 9) RT-773( )/GRC-103(V) | | | | |
| Panel, Distribution (9A1) | 3.10 | 4.19.30 | | |
| Point-to-Point Resistance | | | 1% | |
| Panel, Monitor (9A2) | 3.10 | 4.19.31 | | |
| Point-to-Point Resistance | | | 1% | |



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TABLE VIII - GROUP A INSPECTION (Con't)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Converter, Telephone Signal (9A3) | 3.10 | 4.19.32 | | |
| Oscillator Frequency<br>Signal Receiver Selectivity<br>Signal Receiver Sensitivity | | | 1%<br>FOR<br>THE<br>GROUP | |
| Amplifier Assembly (9A4) | 3.10 | 4.19.33 | | |
| Thru Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Crosstalk | | | 1%<br>FOR<br>THE<br>GROUP | |
| Insert Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Signal Compression | | | 1%<br>FOR<br>THE<br>GROUP | |
| Drop Paths<br>Power Gain<br>Frequency Response<br>Harmonic Distortion<br>Sidetone Level | | | 1%<br>FOR<br>THE<br>GROUP | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Electrical (Main Assemblies) | | | | |
| Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4319( )/GRC-103(V) | | | | |
| Demodulation | 3.4.2.3, 3.4.3.8, 3.4.4.2 | 4.15 | | |
| Noise Figure | 3.4.2.2 | 4.15 | 1% | |
| I.F. Bandwidth | 3.4.2.5 | 4.19.1 | FOR | |
| Received Signal Monitoring | 3.4.2.4 | 4.5 | THE | |
| Receiver Protection | 3.4.2.6 | 4.19.2 | GROUP | |
| Frequency Response | 3.4.2.1 | 4.15 | | |
| VSWR (Duplexers) | 3.4.3.11 | 4.19.19 | | |
| Transmitter, Radio T-9 3( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V) | | | | |
| Frequency Response | 3.4.1.5 | 4.16 | | |
| Power Output | 3.4.1.1 | 4.16 | 1% | |
| Power Dissipation | 3.4.1.3 | 4.16 | FOR | |
| Frequency Setability | 3.4.1.4 | 4.16 | THE | |
| Low Power Alarm | 3.4.1.2 | 4.16 | GROUP | |
| Modulation | 3.4.1.7 | 4.19.3 | | |



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TABLE VIII - GROUP A INSPECTION (Cont'd)

| Inspection | Req. Para. | Insp. Para. | AQL Major | AQL Minor |
|---|---|---|---|---|
| Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) | | | | |
| Tone Generator<br>Deviation<br>Sidetone<br>Peak Limiting | 3.4.4.1.1<br>3.4.4.3<br>3.4.4.11<br>3.4.4.5 | 4.17<br>4.19.3<br>4.19.33<br>4.19.33 | 1% FOR THE GROUP | |
| Antenna AS-3047( )/GRC-103(V) | | | | |
| VSWR<br>Impedance | 3.4.5.5<br>3.4.5.6 | 4.13<br>4.13 | 1% FOR THE GROUP | |
| Electrical (System) | | | | |
| Radio Set AN/GRC-103(V)4 | | | | |
| Operational | 3.17 | 4.10 | 1% | |



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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IX and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B inspection AQL shall be as shown in Table IX and the inspection level shall be S-3 of MIL-STD-105.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in the order listed in Table IX.



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TABLE IX - GROUP B INSPECTION

| Inspection | Req. Para. | Insp. Para. | AQL Major |
|---|---|---|---|
| Electrical (Main Assemblies) | | | |
| Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4319( )/GRC-103(V) | | | |
| Output Impedance | 3.4.3.2 | 4.19.1 | 6.5% FOR |
| Squelch and Alarm | 3.4.3.7.3 | 4.19.1 | THE GROUP |
| Transmitter, Radio T-983( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V) | | | |
| Input Impedance | 3.4.3.1 | 4.19.3 | 6.5% FOR |
| PCM Modulation | 3.4.3.4 | 4.19.3 | THE GROUP |
| Electrical (System) | | | |
| Radio Set AN/GRC-103(V)4 | | | |
| Frequency Response | 3.4.3.3 | 4.18 | |
| PCM Signal-to-Noise | 3.4.3.5 | 4.18 | |
| PCM Distortion | 3.4.3.6 | 4.18 | |
| Pulse Regeneration | 3.4.3.7.1 | 4.18 | 6.5% FOR |
| Timing Signal | 3.4.3.7.2 | 4.18 | THE GROUP |
| Order Wire Distortion | 3.4.4.8 | 4.18 | |
| Order Wire Signal-to-Noise | 3.4.4.4 | 4.18 | |
| Power Requirement | 3.8 | 4.11,4.18 | |
| Duplexer | 3.4.3.9 | 4.19.19 | |
| Duplexer Discrimination | 3.4.3.10 | 4.19.19 | |
| Interchangeability | | | |
| Mechanical | 3.9 | 4.21.1 | 6.5% FOR |
| Electrical | 3.10 | 4.21.2 | THE GROUP |



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4.3.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspections. The inspection shall consist of the inspections specified in Table X. Samples shall be selected in accordance with 4.3.3.1.

TABLE X - GROUP C INSPECTION

| Inspection | Req. Para. | Insp. Para. |
|---|---|---|
| Sub-Group I | | |
| Antenna | | |
| Symmetry | 3.4.5.1 | 4.13 |
| Squint | 3.4.5.2 | 4.13 |
| Minor lobes | 3.4.5.3 | 4.13 |
| Gain | 3.4.5.4 | 4.13 |
| Mast | | |
| Erection | 3.5.1 | 4.14 |
| Operation | 3.5.2 | 4.14 |
| Order wire frequency response | 3.4.4.6 | 4.22 |
| Order wire net loss | 3.4.4.10 | 4.22 |
| Duplex insertion loss | 3.4.3.9 | 4.19.5 |
| Systems level stability | 3.4.3.12 | 4.9.1 |
| Shock, bench-handling | 3.11.4 | 4.7.3 |
| Bounce test - vehicular | 3.11.4 | 4.7.1 |
| Sub-Group II | | |
| Low temperature | 3.11.1.1 | 4.7.4.1 |
| High temperature | 3.11.1.2 | 4.7.4.2 |
| Altitude | 3.11.3 | 4.7.6 |
| Humidity | 3.11.2 | 4.7.5 |
| Sub-Group III | | |
| Electromagnetic interference | 3.4.1.6<br>3.4.2.7<br>3.12 | 4.23 |
| Shock test, ballistic | 3.11.4 | 4.7.2 |



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4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Sub-Group I. For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random from each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group I in the order specified.

4.5.3.1.2 Sub-Group II. For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random for each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group II in the order specified.

4.5.3.1.3 Sub-Group III. For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 500 units. For subsequent Group C inspection,one unit shall be selected at random for each 500 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group III in the order specified.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Group C failures. Actions relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D inspection. This inspection shall consist of the test specified in Table XI and shall be performed on units from lots which have been subjected to and met Group A and B inspection.



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TABLE VI - GROUP D INSPECTION

| Inspection | Requirement Paragraph | Inspection Paragraph |
|---|---|---|
| Reliability | 3.16 | 4.12 |

4.5.5.1 Sampling for Group D inspection. Samples shall be selected at random in accordance with 4.12 and Appendix I. For each production lot, successful completion of the reliability testing must be determined before delivery.

4.5.5.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.5.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall



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not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. All equipment delivered on the contract or order shall be subjected to the requirements of this paragraph. If an equipment fails any test because of the bounce preconditioning, the defect applies to that test.

4.6.2 Electrical preconditioning (burn-in/debugging).

4.6.2.1 Sequence of testing. Before subjecting equipments to this procedure, the contractor must have evidence of compliance with Group A requirements.

4.6.2.2 Test conditions. Each equipment shall be subjected to a 48-hour burn-in period. The last 24 hours of burn-in shall be failure free. The burn-in shall be conducted under the same conditions used for reliability testing (see 4.12). The failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF.

4.6.2.3 Compliance. Units passing burn-in/debugging shall be submitted to Group A and B inspection in that order without benefit of repair, cleaning, realignment, or other changes, unless otherwise specified herein. If after successful completion of burn-in/debugging, the equipment fails either the Group A or B tests, the equipment shall be reworked and resubmitted to burn-in/debugging for a 24-hour failure-free period.

4.7 Service conditions. (Radio only: for antenna test procedures, see SM-A-794072 and for Mast, see SM-A-696070).



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4.7.1 Bounce test; class L units. The equipment shall be tested in accordance with Method 514, Procedure IX, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII. During Part 1, all connections (such as cable assemblies) shall be in place. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.7.2 Shock test; ballistic. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissable. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

4.7.3 Shock test; bench-handling. The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to and after testing, the equipment shall meet full specification performance for the measurements specified in Table VII.



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4.7.4 Temperature test.

4.7.4.1 Low temperature. The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -65°F and the operating temperature shall be -35°F. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.4.2 High temperature. The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 125°F. Prior to, during and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.5 Humidity (moisture-resistance) test. The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.



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4.7.6 Altitude (elevation) test. The equipment shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810, except the chamber internal pressure in Step 2 shall be 282.0 mm of Hg (11.1 in of Hg. or 25,000 ft. above sea level). During and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient atmospheric pressure for frequency changes.

4.8 Standard inspection conditions. Unless otherwise specified, all tests shall be performed at 115 volts +5 percent from a single-phase source operating between 47.5 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure and humidity unless otherwise specified.

4.9 Test specifications (service conditions). The equipment shall be tested in accordance with the following test specifications and meet the requirements of this specification. The tests shall be performed during periods as specified in MIL-STD-810.

4.9.1 Temperature and humidity. 4.2 of SM-A-794148, except the test be performed at channels 2300, 2450, 2600, 2750, 2900 and 3299 instead of those specified. Tests at line frequencies of 47.5Hz and 420Hz, and high and low voltage, shall be performed at any one channel. 4.1 of SM-A-794146 with the exception that tests be performed at channels 2300, 2450, 2600, 2750, 2900 and 3299 instead of those specified. 4.6 of SM-A-794148, except test be performed at one channel only. 4.5 of SM-A-794147, except tests be performed at channels 2300, 2750 and 3299 instead of those specified. 4.1 of SM-A-794147, except test be performed at channel 2600 instead of that specified. 4.2 of SM-A-794146, except frequencies below 10 kHz be deleted from frequency response measurement. 4.4 of SM-A-794146. 4.5.6 of SM-A-794146. 4.3.4 of SM-A-794146.



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4.9.2 Altitude. 4.2 of SM-A-794148, except that the test be performed at channels 2400 and 2900 instead of those specified. 4.6 of SM-A-794148, except that the test be performed at channel 3299 instead of the channels specified. 4.5 of SM-A-794147, except that the test be performed at channel 2700 instead of the channels specified. 4.4 of SM-A-794146.

4.10 Operational test. With the equipment set up in a system, an operational test shall be performed. The test shall consist of voice communications, order wire ringing through the systems plus a visual check of the metering facilities for readings specified in 4.1 of SM-A-794146. (See 3.17).

4.11. A.C. power line, voltage and frequency tests. The following tests shall be performed at room ambient and at channels 2000, 2680 and 3299. The tests shall be performed at 103.5 volts 47.5 Hz, 115 volts 47.5 Hz, 126.5 volts 47.5 Hz, 103.5 volts 420 Hz, 115 volts 420 Hz and 126.5 volts 420 Hz.

| Inspection | Requirement Paragraph |
|---|---|
| 4.2 of SM-A-794148 | 3.4.1 |
| 4.5 of SM-A-794147 | 3.4.2.2 |
| 4.2.5 of SM-A-698826 | 3.4.3.7 |
| 4.2.2 of SM-A-794146 | 3.4.3.4 |
| 4.10 of this specification | 3.4.4.1 |



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4.12 Reliability testing.

4.12.1 Reliability qualification test. The contractor shall perform a Reliability Qualification Test in accordance with Test Plan IV of MIL-STD-781 and Appendix I to this specification. The minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

4.12.2 Production reliability sampling. After successfully passing the reliability qualification test, production reliability sampling shall begin. The production sampling tests shall commence with the first lot available after the models used for first article testing. The samples shall be tested in accordance with Test Plan IV of MIL-STD-781 and Appendix I to this specification. The specified MTBF ($\theta_0$) shall be 1200 hours. The minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

4.12.3 Testing. All testing shall be conducted under the environment of Test Level B of MIL-STD-781. Equipment on-off cycling shall be on for 144 hours, off 8 hours. Vibration shall be for two minutes of every operating hour during one eight-hour period each week.

4.12.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractor's approved test procedures. Paragraph 5.5 through 5.10 inclusive of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.4.8 of MIL-STD-781.

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4.12.5 Rework and retest provisions. Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the lot unless positive action consisting of design change, modification, or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.13 Antenna test procedure. The antenna shall be tested in accordance with SM-A-794072 and meet the requirements of 3.4.5.

4.14 Mast test procedure. The mast shall be tested in accordance with SM-A-696070 and meet the requirements of 3.5.

4.15 Radio Set AN/GRC-103(V)4 receiver section test. The receiver section shall be tested in accordance with paragraphs 4.4, 4.5, 4.3 and 4.2 of SM-A-794147 and meet the requirements of 3.4.2.1, 3.4.2.2, 3.4.2.3, 3.4.3.8, 3.4.4.2 and 3.4.2.4 respectively.

4.16 Radio Set AN/GRC-103(V)4 transmitter section test. The transmitter section shall be tested in accordance with paragraphs 4.2, 4.3, 4.4 and 4.6 of SM-A-794148 and meet the requirements of 3.4.1.4, 3.4.1.1, 3.4.1.2, 3.4.1.3 and 3.4.1.5 respectively.

4.17 Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) test. The receiver-transmitter, order wire shall be tested in accordance with 4.2.5 of SM-A-698807 and meet the requirement of 3.4.4.1.1.



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4.18 Radio Set AN/GRC-103(V)4 test. The radio set shall be tested in accordance with paragraphs 4.2, 4.3, 4.4, 4.5 and 4.1 of SM-A-794146 and meet the requirements of 3.4.3.3, 3.4.3.5, 3.4.3.6, 3.4.3.7.1 and 3.4.3.7.2, 3.4.4.8, 3.4.4.4, 3.4.2.6 and 3.8 respectively.

4.19 Component test. The assemblies, subassemblies and modules of Radio Set AN/GRC-103(V)4 shall be tested in accordance with the following drawings in order to demonstrate compliance with 3.10 of this specification. Specific designations of test frequencies specified herein and/or contractor's prepared test procedures shall be construed as examples to show minimum number of frequencies selected to determine compliance and shall not be construed as restricting testing at other than the listed frequencies. Each requirement shall apply equally to all frequencies within the frequency range. Testing shall be performed on designated frequencies plus six (6) randomly selected frequencies spaced across the band as selected by the contracting officer's technical representative.

4.19.1 Receiver, Radio R-1329( )(P)/GRC-103(V). The Receiver, Radio R-1329( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698826 and meet the requirements thereof. Meeting paragraph 4.2.15 of this drawing demonstrates compliance with 3.4.2.5. Tests for 3.4.3.2 and 3.4.3.7.3 are found in paragraphs 4.2.12 and 4.2.8 of SM-A-698826 respectively.

4.19.2 Amplifier-Converter AM-4319/GRC-103(V). The Amplifier-Converter AM-4319( )/GRC-103(V) shall be tested in accordance with SM-A-794127 and meet the requirements thereof.



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4.19.3 Transmitter, Radio T-983( )(P)/GRC-103(V). The Transmitter, Radio T-983( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698831 and meet the requirements thereof. Tests for the requirements of paragraphs 3.4.4.3, 3.4.3.1, 3.4.3.4 and 3.4.1.7 are found in paragraphs 4.3.5, 4.9, 4.3.1 through 4.3.3, and 4.3.4 of SM-A-698831.

4.19.4 Amplifier-Frequency Multiplier AM-4323/GRC-103(V). The Amplifier-Frequency Multiplier AM-4323( )/GRC-103(V) shall be tested in accordance with SM-A-794156 and meet the requirements thereof.

4.19.5 Synthesizer (1RE1A2/5TR1A2). The synthesizer shall be tested in accordance with SM-A-865041 and meet the requirements thereof.

4.19.6 Restorer, pulse form (1RE1A3). The restorer, pulse form shall be tested in accordance with SM-A-698241 and meet the requirements thereof.

4.19.7 Limiter-discriminator (1RE1A4). The limiter-discriminator shall be tested in accordance with SM-A-698031 and meet the requirements thereof.

4.19.8 Panel,monitor (1RE1A1). The panel, monitor shall be tested in accordance with SM-A-698317 and meet the requirements thereof.

4.19.9 Power supply (1RE1PS1). The power supply shall be tested in accordance with SM-A-698372 and meet the requirements thereof.



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4.19.10 Amplifier-frequency multiplier (1RE1A5). The amplifier-frequency multiplier shall be tested in accordance with SM-A-698610 and meet the requirements thereof.

4.19.11 Amplifier, video (1RE1AR1). The amplifier, video shall be tested in accordance with SM-A-698035 and meet the requirements thereof.

4.19.12 Amplifier, intermediate frequency (1RE1AR2). The amplifier, intermediate frequency shall be tested in accordance with SM-A-698002 and meet the requirements thereof.

4.19.13 Distribution box (1A2A1). The distribution box shall be tested for continuity in accordance with SM-A-698685 and meet the requirements thereof.

4.19.14 Converter, frequency, electronic (39A2). The converter, frequency, electronic shall be tested in accordance with SM-A-794289 and meet the requirements thereof.

4.19.15 Frequency multiplier (39A3). The frequency multiplier shall be tested in accordance with SM-A-794315 and meet the requirements thereof.

4.19.16 Control-indicator (39A4, 40A4). The control-indicator shall be tested in accordance with SM-A-794410 and meet the requirements thereof.



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4.19.17 Amplifier, radio frequency (39AR1). The amplifier, radio frequency shall be tested in accordance with SM-A-794274 and meet the requirements thereof.

4.19.18 Amplifier, intermediate frequency (39AR2). The amplifier, intermediate frequency shall be tested in accordance with SM-A-698644 and meet the requirements thereof.

4.19.19 Filter, bandpass (39FL1, 39FL2, 40FL1). The filter, bandpass shall be tested in accordance with SM-A-794152 and meet the requirements thereof.

4.19.20 Filter, bandpass (39FL3). The filter, bandpass shall be tested in accordance with SM-A-794195 and meet the requirements thereof.

4.19.21 Panel, monitor (5TR1A1). The panel, monitor shall be tested in accordance with SM-A-698790 and meet the requirements thereof.

4.19.22 Control,alarm (5TR1A3). The control, alarm shall be tested in accordance with SM-A-698776 and meet the requirements thereof.

4.19.23 Amplifier-frequency multiplier (5TR1A4). The amplifier-frequency multiplier shall be tested in accordance with SM-A-696443 and meet the requirements thereof.

4.19.24 Amplifier-monitor (5TR1A5). The amplifier-monitor shall be tested in accordance with SM-A-698664 and meet the requirements thereof.



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4.19.25 Power supply (5TR1PS1). The power supply shall be tested in accordance with SM-A-698363 and meet the requirements thereof.

4.19.26 Distribution box (5A2A1). The distribution box shall be tested for continuity in accordance with SM-A-698789 and meet the requirements thereof.

4.19.27 Amplifier-frequency multiplier (40A2). The amplifier-frequency multiplier shall be tested in accordance with SM-A-794356 and meet the requirements thereof.

4.19.28 Regulator, voltage (40A3). The regulator, voltage shall be tested in accordance with SM-A-710010 and meet the requirements thereof.

4.19.29 Amplifier, radio frequency (40AR1). The amplifier, radio frequency shall be tested in accordance with SM-A-794205 and meet the requirements thereof.

4.19.30 Panel, distribution (9A1). The panel, distribution shall be tested in accordance with SM-A-698759 and meet the requirements thereof.

4.19.31 Panel, monitor (9A2). The panel, monitor shall be tested in accordance with SM-A-698695 and meet the requirements thereof.

4.19.32 Converter, telephone signal (9A3). The converter, telephone signal shall be tested in accordance with SM-A-698805 and meet the requirements thereof.



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4.19.33 Amplifier assembly (9A4). The amplifier assembly shall be tested in accordance with SM-A-698806 and meet the requirements thereof. Tests for the requirements of 3.4.4.11 and 3.4.4.5 are found in paragraphs 4.1.2, 4.1.5, 4.2.2, 4.3.2 and 4.3.5, and 4.2.5 of SM-A-698806.

4.20 Visual and mechanical inspection. Parts and equipment shall be examined for the defects listed in MIL-STD-252. In addition, the equipment shall be visually examined for compliance with the requirements of 3.15.

4.21 Interchangeability inspection.

4.21.1 Mechanical interchangeability. Mechanical interchangeability shall be demonstrated by disassembling all normally removable modules from units that have been subjected to and met all other Group B tests except electrical interchangeability, interchanging approximately one-half of the modules (while assuring that one pair of each type module is interchanged) and reassembling. The main frame assemblies (fixed and tuning heads) shall be interchanged between cases. The units shall meet the requirements of 3.9.

4.21.2 Electrical interchangeability. Electrical interchangeability shall be demonstrated by disassembling all normally removable modules from Group B sample units that have been subjected to and met all other Group B tests, interchanging approximately one-half of the modules (while assuring that one pair of each type module is interchanged) and reassembling. The main frame assemblies (fixed and tuning heads) shall be interchanged between cases. The units shall then be operated and measurements made of the transmitter power output and the receiver noise quieting at two frequencies in the band. The units shall meet the requirements of 3.4.1.1 and 3.4.2.2.



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4.22 Order wire operational tests. Two terminals shall be set up to operate in a full duplex PCM system connected through sufficient attenuation. Frequency response and net loss measurements shall be made with the equipment meeting the requirements of 3.4.4.6 and 3.4.4.10.

4.23 Electromagnetic interference tests. The Radio Set AN/GRC-103(V)4 shall be tested for compliance with the electromagnetic interference requirements of 3.4.1.6, 3.4.2.7 and 3.12. Test methods and setups shall be in accordance with MIL-STD-462, Notice 3. All tests except CE06 shall be performed with the equipment operating in a full duplex mode.

4.23.1 Bonding and grounding tests. Prior to the start of EMI tests compliance with the requirement of 3.12.1 shall be verified. Bonding data shall be included in the EMI test report.

4.24 Quality conformance inspection of preparation for delivery. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with MIL-R-45292.



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## 5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery provisions shall be in accordance with MIL-R-45292.

## 6. NOTE

6.1 Intended use. The equipment covered by this specification is a complete, compact, rugged, radio relay set which may be used either as a terminal or modulation frequency patch repeater for intermediate or forward area systems. The radio set is used to provide simultaneous two-way (duplex) transmission. The system provides radio-frequency (rf) channel assignments from 1350 MHz to 1850 MHz. When used in combination, either as a repeater or as a terminal, it is possible to tune and operate the receiver and transmitter on independent frequencies over the frequency range.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see Section 5).

c. When rough handling is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature, tools and running spare parts.

g. First article inspection (see 3.1).



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6.3 Color. The color chip furnished by the procuring agency will be color chip No. 24087 and may be obtained upon request to Commander, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Location of operational tests. It is desirable that operational system tests be performed at a location that will minimize handling after these tests are completed. (See 4.10).

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# APPENDIX I

## RELIABILITY TEST PROGRAM FOR RADIO SET AN/GRC-103(V)4

### 10. SCOPE

10.1 The reliability test program for the Radio Set AN/GRC-103(V)4 shall be completed as per the following test plans:

a. Reliability Qualification Test, in accordance with MIL-STD-781, Test Plan IV, on four (4) systems to be selected from the First Article equipments.

b. Production Reliability Sampling Test, in accordance with MIL-STD-781, Test Plan IV, on two (2), four (4) or six (6) production systems selected from every other month's production lot. (See 30.1).

10.2 The Reliability Demonstration shall be completed after reaching the acceptance criteria of Test Plan IV of MIL-STD-781. The MTBF ($\theta o$) for the above plan shall be 1200 hours for the system.

### 20. TEST CONDITIONS

20.1 The equipments shall be set up in a two-way, two-hop system as shown in Figure 20.1 using suitable attenuators to absorb the output power from each transmitter and to provide an appropriate level of signal to each following receiver. The equipments shall carry a P.C.M. signal and any deterioration in performance of any piece of equipment in the system shall produce error signals at the receiving end. The faulty equipment shall then be located by checking individual meter readings.



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20.2 The test shall be conducted in a suitable test chamber maintained at 40° ±5°C. (95°F to 113°F).

20.3 The equipment shall be accessible to an operator for switching and retuning as required.

20.4 The radio equipment shall operate continuously for periods of 144 hours and shall be switched off for periods of eight hours. A time recording meter shall be used to show total running time for the equipment.

20.5 For two minutes of every operating hour during one eight hour period each week, each equipment shall be vibrated at 2.2g ±10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment.

20.6 The equipment undergoing reliability testing shall be subjected to a burn-in period of at least 48 hours prior to the start of reliability testing. This period of equipment operation shall be performed in accordance with applicable portions of the equipment specification.

30. TEST PROCEDURE

30.1 With the radio sets connected as shown in Figure 20.1, set them to the first group of channels shown in Table 30.1.



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30.2 With the P.C.M. test set on, adjust the transmitter No. A input control to provide ±180 kHz peak deviation as indicated by the panel meter reading (in the green band when switched to 12 CH. P.C.M.).

30.3 Check the meter reading of the following receiver No. B when switched to 12 CH. P.C.M. It should show a reading in the green band.

30.4 Adjust the input control of transmitter No. C for ±180 kHz peak deviation as indicated on the panel meter and check the following receiver No. D.

30.5 Repeat for each equipment group in the "forward" path and then back on the "return" path. Under normal conditions there shall be no appreciable error signal output from the P.C.M. test set and the electronic counter shall only give random indications. In the event of a fault developing anywhere in the system, error signals shall appear at the output and shall give visual indications on the counter. In order to provide continuous unattended monitoring, it is desirable that a digital recorder (summation type) be connected to the output of the counter.

30.6 Every four hours through the test period, the operating channels shall be changed in accordance with the sequence in Table 30.1. If six units are used for production reliability tests, repeat columns of Table 30.1 titled "T1/R2" and "T2/R1" using "T5/R6" and "T6/R5" as the new designations. Channel changes shall continue until an Accept/Reject decision is reached (see MIL-STD-781, Table IV).



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30.7 The meter readings of all equipments shall be logged together with information on channels used after each frequency change. This log shall be in the form shown in Figure 30.1. Order wire signalling shall be checked daily between all equipments on test.

40. FAILURE ACTION

40.1 Only failures occurring within the equipment under test directly as a result of normal operation shall be considered as relevant. Failures in the following categories are classed as non-relevant:

a. Failures in any of the test equipment used.

b. Failures due to accidents or improper test procedures.

c. Failures occurring during the burn-in period.

d. Secondary failures.

40.2 When a failure occurs in any equipment, the failed equipment shall be removed from test, repaired and restored to test.

40.3 The failed equipment shall be appropriately tagged to show the nature of the failure and the accumulated hours of operation. A tag shall be retained on the equipment and a notation of time of return to test added. Full details of the failure shall be recorded as detailed in section 50 and the failure shall be recorded on the form shown in Figure 30.2.



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## 50. TEST RECORDS

50.1 A complete operating log shall be kept during the test to record all pertinent information such as operating hours, off periods, vibration periods, channel changes, meter readings, and times at which failures occur or when units are restored to test.

50.2 In addition to this operating log, an equipment failure record shall be kept for every equipment under test in the format given in MIL-STD-781.

50.3 To assist in the demonstration of the MTBF for the equipment, a failure summary record shall be compiled as shown in MIL-STD-781 and shall include all failures classed as relevant on all equipments under test.



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TABLE 20.1

LIST OF TEST EQUIPMENT

| DESCRIPTION | MANUFACTURER | TYPE |
|---|---|---|
| Vibration Machine 0-60 Hz | L.A.B. Corporation | Model 30-300 or equivalent |
| Accelerometer | Columbia Labs | Model 1700 or equivalent |
| Random PCM Generator | | USAECOM Dwg SM-E-698042 or equivalent |
| Electronic Counter | Hewlett-Packard | 5245L or equivalent |
| Frequency Converter | Hewlett-Packard | 5254A/B or equivalent |
| Directional Coupler | Hewlett-Packard | Model 778D or equivalent |
| Attenuator 60 dB, 2W | Hewlett-Packard | 8491A-660 or equivalent |



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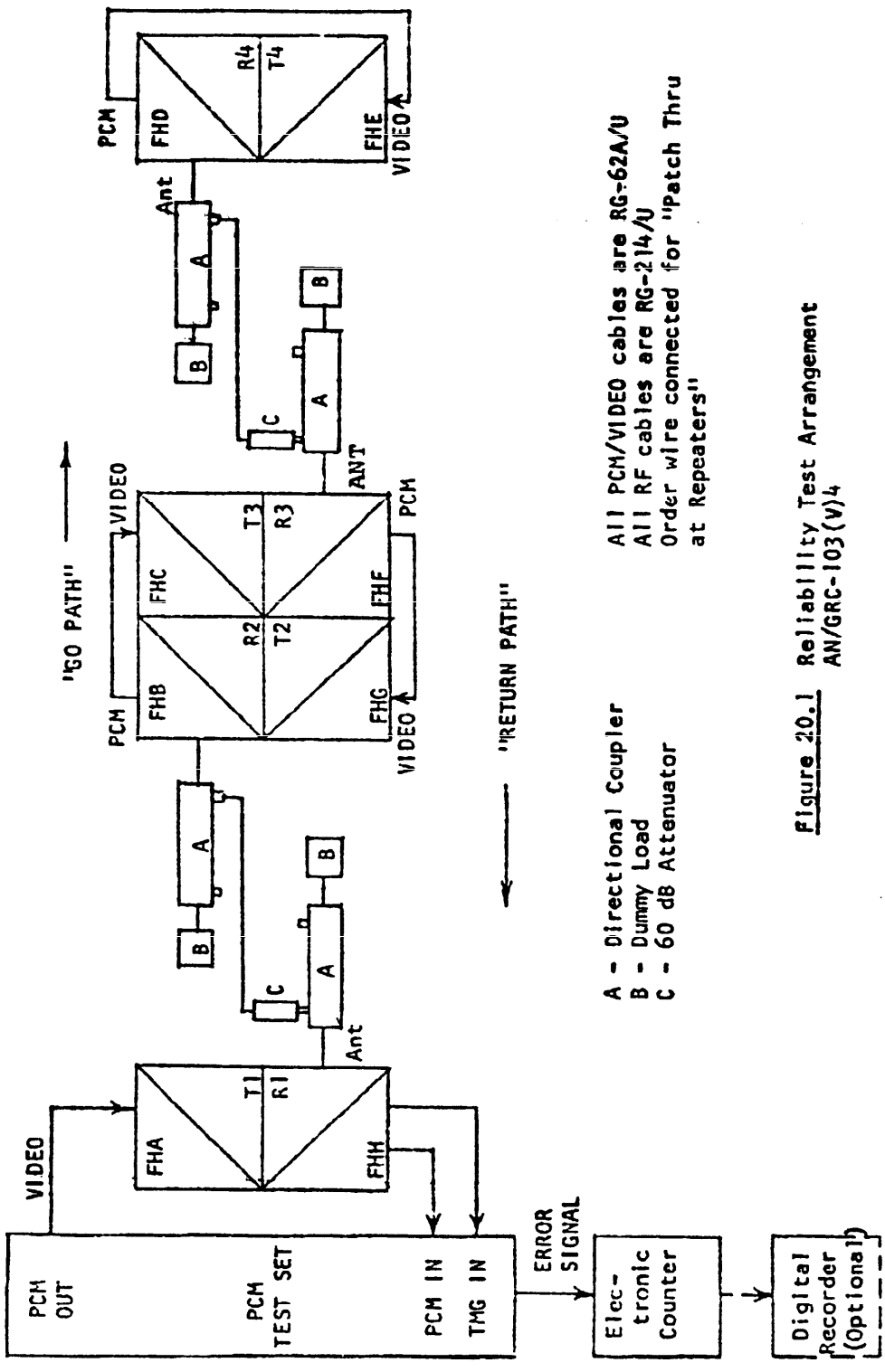



Figure 20.1 Reliability Test Arrangement AN/GRC-103(V)4

NOTE: If two (2) units are on test, remove T3/R3 and T4/R4 and connect the PCM of R2 to the video of T2.

If six (6) units are on test, add T5/R5 and T6/R6 making the following connections: (1) connect the antenna of R5 to the antenna of R6 with the applicable coupler, (2) connect the PCM of R4 to the video of T5, (3) connect the PCM of R5 to the video of T4, and (4) connect the PCM of R6 to the video of T6.



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SCHEDULE OF CHANNEL BAND 4 RELIABILITY TESTS

| Elapsed Hrs. of Operating Time | RF TUNING T1/R2 Chanl. | Freq. MHz | HEAD CHANGE T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | PAIRS T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| Ø | 23ØØ | 135Ø | 25ØØ | 145Ø | 27ØØ | 155Ø | 29ØØ | 165Ø |
| 4 | 31ØØ | 175Ø | 23Ø1 | 135Ø.5 | 25Ø1 | 145Ø.5 | 27Ø1 | 155Ø.5 |
| 8 | 29Ø1 | 165Ø.5 | 31Ø1 | 175Ø.5 | 23Ø2 | 1351 | 25Ø2 | 1451 |
| 12 | 27Ø2 | 1551 | 29Ø2 | 1651 | 31Ø2 | 1751 | 2303 | 1351.5 |
| 16 | 25Ø3 | 1451.5 | 27Ø3 | 1551.5 | 29Ø3 | 1651.5 | 31Ø3 | 1751.5 |
| 2Ø | 23Ø4 | 1352 | 25Ø4 | 1452 | 27Ø4 | 1552 | 29Ø4 | 1652 |
| 24 | 31Ø4 | 1752 | 23Ø5 | 1352.5 | 25Ø5 | 1452.5 | 2705 | 1552.5 |
| 28 | 29Ø5 | 1652.5 | 31Ø5 | 1752.5 | 23Ø6 | 1353 | 25Ø6 | 1453 |
| 32 | 27Ø6 | 1553 | 29Ø6 | 1653 | 31Ø6 | 1753 | 23Ø7 | 1353.5 |
| 36 | 25Ø7 | 1453.5 | 27Ø7 | 1553.5 | 29Ø7 | 1653.5 | 31Ø7 | 1753.5 |
| 40 | 23Ø8 | 1354 | 25Ø8 | 1454 | 27Ø8 | 1554 | 29Ø8 | 1654 |
| 44 | 31Ø8 | 1754 | 23Ø9 | 1354.5 | 25Ø9 | 1454.5 | 27Ø9 | 1554.5 |
| 48 | 29Ø9 | 1654.5 | 31Ø9 | 1754.5 | 2310 | 1355 | 2510 | 1455 |
| 52 | 2710 | 1555 | 2910 | 1655 | 3110 | 1755 | 2311 | 1355.5 |
| 56 | 2511 | 1455.5 | 2711 | 1555.5 | 2911 | 1655.5 | 3111 | 1755.5 |
| 60 | 2312 | 1356 | 2512 | 1456 | 2712 | 1556 | 2912 | 1656 |
| 64 | 3112 | 1756 | 2313 | 1356.5 | 2513 | 1456.5 | 2713 | 1556.5 |
| 68 | 2913 | 1656.5 | 3113 | 1756.5 | 2314 | 1357 | 2514 | 1457 |
| 72 | 2714 | 1557 | 2914 | 1657 | 3114 | 1757 | 2315 | 1357.5 |
| 76 | 2515 | 1457.5 | 2715 | 1557.5 | 2915 | 1657.5 | 3115 | 1757.5 |
| 80 | 2316 | 1358 | 2516 | 1458 | 2716 | 1558 | 2916 | 1658 |
| 84 | 3116 | 1758 | 2317 | 1358.5 | 2517 | 1458.5 | 2717 | 1558.5 |
| 88 | 2917 | 1658.5 | 3117 | 1758.5 | 2318 | 1359 | 2518 | 1459 |
| 92 | 2718 | 1559 | 2918 | 1659 | 3118 | 1759 | 2319 | 1359.5 |
| 96 | 2519 | 1459.5 | 2719 | 1559.5 | 2919 | 1659.5 | 3119 | 1759.5 |
| 1ØØ | 232Ø | 136Ø | 252Ø | 146Ø | 272Ø | 156Ø | 292Ø | 166Ø |
| 1Ø4 | 312Ø | 176Ø | 2321 | 136Ø.5 | 2521 | 146Ø.5 | 2721 | 156Ø.5 |
| 1Ø8 | 2921 | 166Ø.5 | 3121 | 176Ø.5 | 2322 | 1361 | 2522 | 1461 |
| 112 | 2722 | 1561 | 2922 | 1661 | 3122 | 1761 | 2323 | 1361.5 |
| 116 | 2523 | 1461.5 | 2723 | 1561.5 | 2923 | 1661.5 | 3123 | 1761.5 |
| 120 | 2324 | 1362 | 2524 | 1462 | 2724 | 1562 | 2924 | 1662 |

TABLE 30.1



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| Elapsed Hrs. of Operating Time | RT TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 124 | 3124 | 1762 | 2325 | 1362.5 | 2525 | 1462.5 | 2725 | 1562.5 |
| 128 | 2925 | 1662.5 | 3125 | 1762.5 | 2326 | 1363 | 2526 | 1463 |
| 132 | 2726 | 1563 | 2926 | 1663 | 3126 | 1763 | 2327 | 1363.5 |
| 136 | 2727 | 1463.5 | 2727 | 1563.5 | 2927 | 1663.5 | 3127 | 1763.5 |
| 140 | 2328 | 1364 | 2528 | 1464 | 2728 | 1564 | 2928 | 1664 |
| 144 | 3128 | 1764 | 2329 | 1364.5 | 2529 | 1464.5 | 2729 | 1564.5 |
| 148 | 2929 | 1664.5 | 3129 | 1764.5 | 2330 | 1365 | 2530 | 1465 |
| 152 | 2730 | 1565 | 2930 | 1665 | 3130 | 1765 | 2331 | 1365.5 |
| 156 | 2531 | 1465.5 | 2731 | 1565.5 | 2931 | 1665.5 | 3131 | 1765.5 |
| 160 | 2332 | 1366 | 2532 | 1466 | 2732 | 1566 | 2932 | 1666 |
| 164 | 3132 | 1766 | 2333 | 1366.5 | 2533 | 1466.5 | 2733 | 1566.5 |
| 168 | 2933 | 1666.5 | 3133 | 1766.5 | 2334 | 1367 | 2534 | 1467 |
| 172 | 2734 | 1567 | 2934 | 1667 | 3134 | 1767 | 2335 | 1367.5 |
| 176 | 2535 | 1467.5 | 2735 | 1567.5 | 2935 | 1667.5 | 3135 | 1767.5 |
| 180 | 2336 | 1368 | 2536 | 1468 | 2736 | 1568 | 2936 | 1668 |
| 184 | 3136 | 1768 | 2337 | 1368.5 | 2537 | 1468.5 | 2737 | 1568.5 |
| 188 | 2937 | 1668.5 | 3137 | 1768.5 | 2338 | 1369 | 2538 | 1469 |
| 192 | 2738 | 1569 | 2938 | 1669 | 3138 | 1769 | 2339 | 1369.5 |
| 196 | 2539 | 1469.5 | 2739 | 1569.5 | 2939 | 1669.5 | 3139 | 1769.5 |
| 200 | 2340 | 1370 | 2540 | 1470 | 2740 | 1570 | 2940 | 1670 |
| 204 | 3140 | 1770 | 2341 | 1370.5 | 2541 | 1470.5 | 2741 | 1570.5 |
| 208 | 2941 | 1670.5 | 3141 | 1770.5 | 2342 | 1371 | 2542 | 1471 |
| 212 | 2742 | 1571 | 2942 | 1671 | 3142 | 1771 | 2343 | 1371.5 |
| 216 | 2543 | 1471.5 | 2743 | 1571.5 | 2943 | 1671.5 | 3143 | 1771.5 |
| 220 | 2344 | 1372 | 2544 | 1472 | 2744 | 1572 | 2944 | 1672 |
| 224 | 3144 | 1772 | 2345 | 1372.5 | 2545 | 1472.5 | 2745 | 1572.5 |
| 228 | 2945 | 1672.5 | 3145 | 1772.5 | 2346 | 1373 | 2546 | 1473 |
| 232 | 2746 | 1573 | 2946 | 1673 | 3146 | 1773 | 2347 | 1373.5 |
| 236 | 2547 | 1473.5 | 2747 | 1573.5 | 2947 | 1673.5 | 3147 | 1773.5 |
| 240 | 2348 | 1374 | 2548 | 1474 | 2748 | 1574 | 2948 | 1674 |

TABLE 30.1 (cont'd)



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| Elapsed Hrs. of Operating Time | RF TUNING T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | HEAD CHANGE T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | PAIRS Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 244 | 3148 | 1774 | 2349 | 1374.5 | 2549 | 1474.5 | 2749 | 1574.5 |
| 248 | 2949 | 1674.5 | 3149 | 1774.5 | 2350 | 1375 | 2550 | 1475 |
| 252 | 2750 | 1575 | 2950 | 1675 | 3150 | 1775 | 2351 | 1375.5 |
| 256 | 2551 | 1475.5 | 2751 | 1575.5 | 2951 | 1675.5 | 3151 | 1775.5 |
| 260 | 2352 | 1376 | 2552 | 1476 | 2752 | 1576 | 2952 | 1676 |
| 264 | 3152 | 1776 | 2353 | 1376.5 | 2553 | 1476.5 | 2753 | 1576.5 |
| 268 | 2953 | 1676.5 | 3153 | 1776.5 | 2354 | 1377 | 2554 | 1477 |
| 272 | 2754 | 1577 | 2954 | 1677 | 3154 | 1777 | 2355 | 1377.5 |
| 276 | 2555 | 1477.5 | 2755 | 1577.5 | 2955 | 1677.5 | 3155 | 1777.5 |
| 280 | 2356 | 1378 | 2556 | 1478 | 2756 | 1578 | 2956 | 1678 |
| 284 | 3156 | 1778 | 2357 | 1378.5 | 2557 | 1478.5 | 2757 | 1578.5 |
| 288 | 2957 | 1678.5 | 3157 | 1778.5 | 2358 | 1379 | 2558 | 1479 |
| 292 | 2758 | 1579 | 2958 | 1679 | 3158 | 1779 | 2359 | 1379.5 |
| 296 | 2559 | 1479.5 | 2759 | 1579.5 | 2959 | 1679.5 | 3159 | 1779.5 |
| 300 | 2360 | 1380 | 2560 | 1480 | 2760 | 1580 | 2960 | 1680 |
| 304 | 3160 | 1780 | 2361 | 1380.5 | 2561 | 1480.5 | 2761 | 1580.5 |
| 308 | 2961 | 1680.5 | 3161 | 1780.5 | 2362 | 1381 | 2562 | 1481 |
| 312 | 2762 | 1581 | 2962 | 1681 | 3162 | 1781 | 2363 | 1381.5 |
| 316 | 2563 | 1481.5 | 2763 | 1581.5 | 2963 | 1681.5 | 3163 | 1781.5 |
| 320 | 2364 | 1382 | 2564 | 1482 | 2764 | 1582 | 2964 | 1682 |
| 324 | 3164 | 1782 | 2365 | 1382.5 | 2565 | 1482.5 | 2765 | 1582.5 |
| 328 | 2965 | 1682.5 | 3165 | 1782.5 | 2366 | 1383 | 2566 | 1483 |
| 332 | 2766 | 1583 | 2966 | 1683 | 3166 | 1783 | 2367 | 1383.5 |
| 336 | 2567 | 1483.5 | 2767 | 1583.5 | 2967 | 1683.5 | 3167 | 1783.5 |
| 340 | 2368 | 1384 | 2568 | 1484 | 2768 | 1584 | 2968 | 1684 |
| 344 | 3168 | 1784 | 2369 | 1384.5 | 2569 | 1484.5 | 2769 | 1584.5 |
| 348 | 2969 | 1684.5 | 3169 | 1784.5 | 2370 | 1385 | 2570 | 1485 |
| 352 | 2770 | 1585 | 2970 | 1685 | 3170 | 1785 | 2371 | 1385.5 |
| 356 | 2571 | 1485.5 | 2771 | 1585.5 | 2971 | 1685.5 | 3171 | 1785.5 |
| 360 | 2372 | 1386 | 2572 | 1486 | 2772 | 1586 | 2972 | 1686 |

TABLE 30.1 (cont'd)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

| Elapsed Hrs. of Operat-ing Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 364 | 3172 | 1786 | 2373 | 1386.5 | 2573 | 1486.5 | 2773 | 1586.5 |
| 368 | 2973 | 1686.5 | 3173 | 1786.5 | 2374 | 1387 | 2574 | 1487 |
| 372 | 2774 | 1587 | 2974 | 1687 | 3174 | 1787 | 2375 | 1387.5 |
| 376 | 2575 | 1487.5 | 2775 | 1587.5 | 2975 | 1687.5 | 3175 | 1787.5 |
| 380 | 2376 | 1388 | 2576 | 1488 | 2776 | 1588 | 2976 | 1688 |
| 384 | 3176 | 1788 | 2377 | 1388.5 | 2577 | 1488.5 | 2777 | 1588.5 |
| 388 | 2977 | 1688.5 | 3177 | 1788.5 | 2378 | 1389 | 2578 | 1489 |
| 392 | 2778 | 1589 | 2978 | 1689 | 3178 | 1789 | 2379 | 1389.5 |
| 396 | 2579 | 1489.5 | 2779 | 1589.5 | 2979 | 1689.5 | 3179 | 1789.5 |
| 400 | 2380 | 1390 | 2580 | 1490 | 2780 | 1590 | 2980 | 1690 |
| 404 | 3180 | 1790 | 2381 | 1390.5 | 2581 | 1490.5 | 2781 | 1590.5 |
| 408 | 2981 | 1690.5 | 3181 | 1790.5 | 2382 | 1391 | 2582 | 1491 |
| 412 | 2782 | 1591 | 2982 | 1691 | 3182 | 1791 | 2383 | 1391.5 |
| 416 | 2583 | 1491.5 | 2783 | 1591.5 | 2983 | 1691.5 | 3183 | 1791.5 |
| 420 | 2384 | 1392 | 2584 | 1492 | 2784 | 1592 | 2984 | 1692 |
| 424 | 3184 | 1792 | 2385 | 1392.5 | 2585 | 1492.5 | 2785 | 1592.5 |
| 428 | 2985 | 1692.5 | 3185 | 1792.5 | 2386 | 1393 | 2586 | 1493 |
| 432 | 2786 | 1593 | 2986 | 1693 | 3186 | 1793 | 2387 | 1393.5 |
| 436 | 2587 | 1493.5 | 2787 | 1593.5 | 2987 | 1693.5 | 3187 | 1793.5 |
| 440 | 2388 | 1394 | 2588 | 1494 | 2788 | 1594 | 2988 | 1694 |
| 444 | 3188 | 1794 | 2389 | 1394.5 | 2589 | 1494.5 | 2789 | 1594.5 |
| 448 | 2989 | 1694.5 | 3189 | 1794.5 | 2390 | 1395 | 2590 | 1495 |
| 452 | 2790 | 1595 | 2990 | 1695 | 3190 | 1795 | 2391 | 1395.5 |
| 456 | 2591 | 1495.5 | 2791 | 1595.5 | 2991 | 1695.5 | 3191 | 1795.5 |
| 460 | 2392 | 1396 | 2592 | 1496 | 2792 | 1596 | 2992 | 1696 |
| 464 | 3192 | 1796 | 2393 | 1396.5 | 2593 | 1496.5 | 2793 | 1596.5 |
| 468 | 2993 | 1696.5 | 3193 | 1796.5 | 2394 | 1397 | 2594 | 1497 |
| 472 | 2794 | 1597 | 2994 | 1697 | 3194 | 1797 | 2395 | 1397.5 |
| 476 | 2595 | 1497.5 | 2795 | 1597.5 | 2995 | 1697.5 | 3195 | 1797.5 |
| 480 | 2396 | 1398 | 2596 | 1498 | 2796 | 1598 | 2996 | 1698 |

TABLE 30.1 (cont'd)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
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| Elapsed Hrs. of Operating Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 484 | 3196 | 1798 | 2397 | 1398.5 | 2597 | 1498.5 | 2797 | 1598.5 |
| 488 | 2997 | 1698.5 | 3197 | 1798.5 | 2398 | 1399 | 2598 | 1499 |
| 492 | 2798 | 1599 | 2998 | 1699 | 3198 | 1799 | 2399 | 1399.5 |
| 496 | 2599 | 1499.5 | 2799 | 1599.5 | 2999 | 1699.5 | 3199 | 1799.5 |
| 500 | 2400 | 1400 | 2600 | 1500 | 2800 | 1600 | 3000 | 1700 |
| 504 | 3200 | 1800 | 2401 | 1400.5 | 2601 | 1500.5 | 2801 | 1600.5 |
| 508 | 3001 | 1700.5 | 3201 | 1800.5 | 2402 | 1401 | 2602 | 1501 |
| 512 | 2802 | 1601 | 3002 | 1701 | 3202 | 1801 | 2403 | 1401.5 |
| 516 | 2603 | 1501.5 | 2803 | 1601.5 | 3003 | 1701.5 | 3203 | 1801.5 |
| 520 | 2404 | 1402 | 2604 | 1502 | 2804 | 1602 | 3004 | 1702 |
| 524 | 3204 | 1802 | 2405 | 1402.5 | 2605 | 1502.5 | 2805 | 1602.5 |
| 528 | 3005 | 1702.5 | 3205 | 1802.5 | 2406 | 1403 | 2606 | 1503 |
| 532 | 2806 | 1603 | 3006 | 1703 | 3206 | 1803 | 2407 | 1403.5 |
| 536 | 2607 | 1503.5 | 2807 | 1603.5 | 3007 | 1703.5 | 3207 | 1803.5 |
| 540 | 2408 | 1404 | 2608 | 1504 | 2808 | 1604 | 3008 | 1704 |
| 544 | 3208 | 1804 | 2409 | 1404.5 | 2609 | 1504.5 | 2809 | 1604.5 |
| 548 | 3009 | 1704.5 | 3209 | 1804.5 | 2410 | 1405 | 2610 | 1505 |
| 552 | 2810 | 1605 | 3010 | 1705 | 3210 | 1805 | 2411 | 1405.5 |
| 556 | 2611 | 1505.5 | 2811 | 1605.5 | 3011 | 1705.5 | 3211 | 1805.5 |
| 560 | 2412 | 1406 | 2612 | 1506 | 2812 | 1606 | 3012 | 1706 |
| 564 | 3212 | 1806 | 2413 | 1406.5 | 2613 | 1506.5 | 2813 | 1606.5 |
| 568 | 3013 | 1706.5 | 3213 | 1806.5 | 2414 | 1407 | 2614 | 1507 |
| 572 | 2814 | 1607 | 3014 | 1707 | 3214 | 1807 | 2415 | 1407.5 |
| 576 | 2615 | 1507.5 | 2815 | 1607.5 | 3015 | 1707.5 | 3215 | 1807.5 |
| 580 | 2416 | 1408 | 2616 | 1508 | 2816 | 1608 | 3016 | 1708 |
| 584 | 3216 | 1808 | 2417 | 1408.5 | 2617 | 1508.5 | 2817 | 1608.5 |
| 588 | 3017 | 1708.5 | 3217 | 1808.5 | 2418 | 1409 | 2618 | 1509 |
| 592 | 2818 | 1609 | 3018 | 1709 | 3218 | 1809 | 2419 | 1409.5 |
| 596 | 2619 | 1509.5 | 2819 | 1609.5 | 3019 | 1709.5 | 3219 | 1809.5 |
| 600 | 2420 | 1410 | 2620 | 1510 | 2820 | 1610 | 3020 | 1710 |

TABLE 30.1 (cont'd)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
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| Elapsed Hrs. of Operating Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 604 | 3220 | 1810 | 2421 | 1410.5 | 2621 | 1510.5 | 2821 | 1610.5 |
| 608 | 3021 | 1710.5 | 3221 | 1810.5 | 2422 | 1411 | 2622 | 1511 |
| 612 | 2822 | 1611 | 3022 | 1711 | 3222 | 1811 | 2423 | 1411.5 |
| 616 | 2623 | 1511.5 | 2823 | 1611.5 | 3023 | 1711.5 | 3223 | 1811.5 |
| 620 | 2424 | 1412 | 2624 | 1512 | 2824 | 1612 | 3024 | 1712 |
| 624 | 3224 | 1812 | 2425 | 1412.5 | 2625 | 1512.5 | 2825 | 1612.5 |
| 628 | 3025 | 1712.5 | 3225 | 1812.5 | 2426 | 1413 | 2626 | 1513 |
| 632 | 2826 | 1613 | 3026 | 1713 | 3226 | 1813 | 2427 | 1413.5 |
| 636 | 2627 | 1513.5 | 2827 | 1613.5 | 3027 | 1713.5 | 3227 | 1813.5 |
| 640 | 2428 | 1414 | 2628 | 1514 | 2828 | 1614 | 3028 | 1714 |
| 644 | 3228 | 1814 | 2429 | 1414.5 | 2629 | 1514.5 | 2829 | 1614.5 |
| 648 | 3029 | 1714.5 | 3229 | 1814.5 | 2430 | 1415 | 2630 | 1515 |
| 652 | 2830 | 1615 | 3030 | 1715 | 3230 | 1815 | 2431 | 1415.5 |
| 656 | 2631 | 1515.5 | 2631 | 1615.5 | 3031 | 1715.5 | 3231 | 1815.5 |
| 660 | 2432 | 1416 | 2632 | 1516 | 2832 | 1616 | 3032 | 1716 |
| 664 | 3232 | 1816 | 2433 | 1416.5 | 2633 | 1516.5 | 2833 | 1616.5 |
| 668 | 3033 | 1716.5 | 3233 | 1816.5 | 2434 | 1417 | 2634 | 1517 |
| 672 | 2834 | 1617 | 3034 | 1717 | 3234 | 1817 | 2435 | 1417.5 |
| 676 | 2635 | 1517.5 | 2835 | 1617.5 | 3035 | 1717.5 | 3235 | 1817.5 |
| 680 | 2436 | 1418 | 2636 | 1518 | 2836 | 1618 | 3036 | 1718 |
| 684 | 3236 | 1818 | 2437 | 1418.5 | 2637 | 1518.5 | 2837 | 1618.5 |
| 688 | 3037 | 1718.5 | 3237 | 1818.5 | 2438 | 1419 | 2638 | 1519 |
| 692 | 2838 | 1619 | 3038 | 1719 | 3238 | 1819 | 2439 | 1419.5 |
| 696 | 2639 | 1519.5 | 2839 | 1619.5 | 3039 | 1719.5 | 3239 | 1819.5 |
| 700 | 2440 | 1420 | 2640 | 1520 | 2840 | 1620 | 3040 | 1720 |
| 704 | 3240 | 1820 | 2441 | 1420.5 | 2641 | 1520.5 | 2841 | 1620.5 |
| 708 | 3041 | 1720.5 | 3241 | 1820.5 | 2442 | 1421 | 2642 | 1521 |
| 712 | 2842 | 1621 | 3042 | 1721 | 3242 | 1821 | 2443 | 1421.5 |
| 716 | 2643 | 1521.5 | 2843 | 1621.5 | 3043 | 1721.5 | 3243 | 1821.5 |
| 720 | 2444 | 1422 | 2644 | 1522 | 2844 | 1622 | 3044 | 1722 |

TABLE 30.1 (cont'd)



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| Elapsed Hrs. of Operating Time | RF TUNING T1/R2 Chanl. | Freq. MHz | HEAD T2/R1 Chanl. | Freq. MHz | CHANGE T3/R4 Chanl. | Freq. MHz | PAIRS T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 724 | 3244 | 1822 | 2445 | 1422.5 | 2645 | 1522.5 | 2845 | 1622.5 |
| 728 | 3045 | 1722.5 | 3245 | 1822.5 | 2446 | 1423 | 2646 | 1523 |
| 732 | 2846 | 1623 | 3046 | 1723 | 3246 | 1623 | 2447 | 1423.5 |
| 736 | 2647 | 1523.5 | 2847 | 1623.5 | 3247 | 1723.5 | 3247 | 1823.5 |
| 740 | 2448 | 1424 | 2648 | 1524 | 2646 | 1624 | 3048 | 1724 |
| 744 | 3248 | 1624 | 2449 | 1424.5 | 2649 | 1524.5 | 2849 | 1624.5 |
| 748 | 3049 | 1724.5 | 3249 | 1824.5 | 2450 | 1425 | 2650 | 1525 |
| 752 | 2850 | 1625 | 3050 | 1725 | 3250 | 1825 | 2451 | 1425.5 |
| 756 | 2651 | 1525.5 | 2851 | 1625.5 | 3051 | 1725.5 | 3251 | 1825.5 |
| 760 | 2452 | 1426 | 2652 | 1526 | 2852 | 1626 | 3052 | 1726 |
| 764 | 3252 | 1826 | 2453 | 1426.5 | 2653 | 1526.5 | 2853 | 1626.5 |
| 768 | 3053 | 1726.5 | 3253 | 1826.5 | 2454 | 1427 | 2654 | 1527 |
| 772 | 2854 | 1627 | 3054 | 1727 | 3254 | 1827 | 2455 | 1427.5 |
| 776 | 2655 | 1527.5 | 2655 | 1627.5 | 3055 | 1727.5 | 3255 | 1827.5 |
| 780 | 2456 | 1428 | 2656 | 1528 | 2856 | 1628 | 3056 | 1728 |
| 784 | 3256 | 1828 | 2457 | 1428.5 | 2657 | 1528.5 | 2857 | 1628.5 |
| 788 | 3057 | 1728.5 | 3257 | 1828.5 | 2458 | 1429 | 2658 | 1529 |
| 792 | 2858 | 1629 | 3058 | 1729 | 3258 | 1829 | 2459 | 1429.5 |
| 796 | 2659 | 1529.5 | 2859 | 1629.5 | 3059 | 1729.5 | 3259 | 1829.5 |
| 800 | 2460 | 1430 | 2660 | 1530 | 2860 | 1630 | 3060 | 1730 |
| 804 | 3260 | 1830 | 2461 | 1430.5 | 2661 | 1530.5 | 2861 | 1630.5 |
| 808 | 3061 | 1730.5 | 3261 | 1830.5 | 2462 | 1431 | 2662 | 1531 |
| 812 | 2862 | 1631 | 3062 | 1731 | 3262 | 1831 | 2463 | 1431.5 |
| 816 | 2663 | 1531.5 | 2863 | 1631.5 | 3063 | 1731.5 | 3263 | 1831.5 |
| 820 | 2464 | 1432 | 2664 | 1532 | 2864 | 1632 | 3064 | 1732 |
| 824 | 3264 | 1832 | 2465 | 1432.5 | 2665 | 1532.5 | 2865 | 1632.5 |
| 828 | 3065 | 1732.5 | 3265 | 1832.5 | 2466 | 1433 | 2666 | 1533 |
| 832 | 2866 | 1633 | 3066 | 1733 | 3266 | 1833 | 2467 | 1433.5 |
| 836 | 2667 | 1533.5 | 2867 | 1633.5 | 3067 | 1733.5 | 3267 | 1833.5 |
| 840 | 2468 | 1434 | 2668 | 1534 | 2868 | 1634 | 3068 | 1734 |

TABLE 30.1 (cont'd)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

| Elapsed Hrs. of Operating Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 844 | 3268 | 1634 | 2469 | 1434.5 | 2869 | 1534.5 | 2869 | 1634.5 |
| 848 | 3069 | 1734.5 | 3269 | 1834.5 | 2470 | 1435 | 2670 | 1535 |
| 852 | 2870 | 1635 | 3070 | 1735 | 3270 | 1835 | 2471 | 1435.5 |
| 856 | 2671 | 1535.5 | 2871 | 1635.5 | 3071 | 1735.5 | 3271 | 1835.5 |
| 860 | 2472 | 1436 | 2672 | 1536 | 2872 | 1636 | 3072 | 1736 |
| 864 | 3272 | 1836 | 2473 | 1436.5 | 2673 | 1536.5 | 2873 | 1636.5 |
| 868 | 3073 | 1736.5 | 3273 | 1836.5 | 2474 | 1437 | 2674 | 1537 |
| 872 | 2874 | 1637 | 3074 | 1737 | 3274 | 1837 | 2475 | 1437.5 |
| 876 | 2675 | 1537.5 | 2875 | 1637.5 | 3075 | 1737.5 | 3275 | 1837.5 |
| 880 | 2476 | 1438 | 2676 | 1538 | 2874 | 1638 | 3276 | 1738 |
| 884 | 3276 | 1838 | 2477 | 1438.5 | 2677 | 1538.5 | 2877 | 1638.5 |
| 888 | 3077 | 1738.5 | 3277 | 1838.5 | 2478 | 1439 | 2678 | 1539 |
| 892 | 2878 | 1639 | 3078 | 1739 | 3278 | 1839 | 2479 | 1439.5 |
| 896 | 2679 | 1539.5 | 2879 | 1639.5 | 3079 | 1739.5 | 3279 | 1839.5 |
| 900 | 2480 | 1440 | 2680 | 1540 | 2880 | 1640 | 3080 | 1740 |
| 904 | 3280 | 1840 | 2481 | 1440.5 | 2681 | 1540.5 | 2881 | 1640.5 |
| 908 | 3081 | 1740.5 | 3281 | 1840.5 | 2482 | 1441 | 2682 | 1541 |
| 912 | 2882 | 1641 | 3082 | 1741 | 3282 | 1841 | 2483 | 1441.5 |
| 916 | 2683 | 1541.5 | 2883 | 1641.5 | 3083 | 1741.5 | 3283 | 1841.5 |
| 920 | 2484 | 1442 | 2684 | 1542 | 2884 | 1642 | 3084 | 1742 |
| 924 | 3284 | 1842 | 2485 | 1442.5 | 2685 | 1542.5 | 2885 | 1642.5 |
| 928 | 3085 | 1742.5 | 3285 | 1842.5 | 2486 | 1443 | 2686 | 1543 |
| 932 | 2886 | 1643 | 3086 | 1743 | 3286 | 1843 | 2487 | 1443.5 |
| 936 | 2687 | 1543.5 | 2887 | 1643.5 | 3087 | 1743.5 | 3287 | 1843.5 |
| 940 | 2488 | 1444 | 2688 | 1544 | 2888 | 1644 | 3088 | 1744 |
| 944 | 3288 | 1844 | 2489 | 1444.5 | 2689 | 1544.5 | 2889 | 1644.5 |
| 948 | 3089 | 1744.5 | 3289 | 1844.5 | 2490 | 1445 | 2690 | 1545 |
| 952 | 2890 | 1645 | 3090 | 1745 | 3290 | 1845 | 2491 | 1445.5 |
| 956 | 2691 | 1545.5 | 2891 | 1645.5 | 3091 | 1745.5 | 3291 | 1845.5 |
| 960 | 2492 | 1446 | 2692 | 1546 | 2892 | 1646 | 3092 | 1746 |

**TABLE 30.1 (cont'd)**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
Check the source to verify that this is the current version before use.

| Elapsed Hrs. of Operating Time | RF TUNING T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | HEAD Freq. MHz | T3/R4 Chanl. | CHANGE Freq. MHz | T4/R3 Chanl. | PAIRS Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 964 | 3292 | 1846 | 2493 | 1446.5 | 2693 | 1546.5 | 2893 | 1646.5 |
| 968 | 3093 | 1746.5 | 3293 | 1646.5 | 2494 | 1447 | 2694 | 1547 |
| 972 | 2694 | 1647 | 3094 | 1747 | 3294 | 1847 | 2495 | 1447.5 |
| 976 | 2695 | 1547.5 | 2895 | 1647.5 | 3095 | 1747.5 | 3295 | 1847.5 |
| 980 | 2496 | 1448 | 2696 | 1548 | 2896 | 1648 | 3096 | 1748 |
| 984 | 3296 | 1848 | 2497 | 1448.5 | 2697 | 1548.5 | 2897 | 1648.5 |
| 988 | 3097 | 1748.5 | 3297 | 1848.5 | 2498 | 1449 | 2698 | 1549 |
| 992 | 2898 | 1649 | 3098 | 1749 | 3298 | 1849 | 2499 | 1449.5 |
| 996 | 2699 | 1549.5 | 2899 | 1649.5 | 3099 | 1749.5 | 3299 | 1849.5 |
| 1000 | 2500 | 1450 | 2700 | 1550 | 2900 | 1650 | 3100 | 1750 |
| 1004 | 2501 | 1450.5 | 2701 | 1550.5 | 2901 | 1650.5 | 3101 | 1750.5 |
| 1008 | 2502 | 1451 | 2702 | 1551 | 2902 | 1651 | 3102 | 1751 |
| 1012 | 2503 | 1451.5 | 2703 | 1551.5 | 2903 | 1651.5 | 3103 | 1751.5 |
| 1016 | 2504 | 1452 | 2704 | 1552 | 2904 | 1652 | 3104 | 1752 |
| 1020 | 2505 | 1452.5 | 2705 | 1552.5 | 2905 | 1652.5 | 3105 | 1752.5 |
| 1024 | 2506 | 1453 | 2706 | 1553 | 2906 | 1653 | 3106 | 1753 |
| 1028 | 2507 | 1453.5 | 2707 | 1553.5 | 2907 | 1653.5 | 3107 | 1753.5 |
| 1032 | 2508 | 1454 | 2708 | 1554 | 2908 | 1654 | 3108 | 1754 |
| 1036 | 2509 | 1454.5 | 2709 | 1554.5 | 2909 | 1654.5 | 3109 | 1754.5 |
| 1040 | 2510 | 1455 | 2710 | 1555 | 2910 | 1655 | 3110 | 1755 |
| 1044 | 2511 | 1455.5 | 2711 | 1555.5 | 2911 | 1655.5 | 3111 | 1755.5 |
| 1048 | 2512 | 1456 | 2712 | 1556 | 2912 | 1656 | 3112 | 1756 |
| 1052 | 2513 | 1456.5 | 2713 | 1556.5 | 2913 | 1656.5 | 3113 | 1756.5 |
| 1056 | 2514 | 1457 | 2714 | 1557 | 2914 | 1657 | 3114 | 1757 |
| 1060 | 2515 | 1457.5 | 2715 | 1557.5 | 2915 | 1657.5 | 3115 | 1757.5 |
| 1064 | 2516 | 1458 | 2716 | 1558 | 2916 | 1658 | 3116 | 1758 |
| 1068 | 2517 | 1458.5 | 2717 | 1558.5 | 2917 | 1658.5 | 3117 | 1758.5 |
| 1072 | 2518 | 1459 | 2718 | 1559 | 2918 | 1659 | 3118 | 1759 |
| 1076 | 2519 | 1459.5 | 2719 | 1559.5 | 2919 | 1659.5 | 3119 | 1759.5 |
| 1080 | 2520 | 1460 | 2720 | 1560 | 2920 | 1660 | 3120 | 1760 |

TABLE 30.1 (cont'd)



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| Elapsed Hrs. of Operating Time | RF TUNING T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | HEAD Freq. MHz | T3/R4 Chanl. | CHANGE Freq. MHz. | T4/R3 Chanl. | PAIRS Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 1084 | 2521 | 1460.5 | 2721 | 1562.5 | 2921 | 1660.5 | 3121 | 1762.5 |
| 1088 | 2522 | 1461 | 2722 | 1563 | 2922 | 1661 | 3122 | 1763 |
| 1092 | 2523 | 1461.5 | 2723 | 1561.5 | 2923 | 1661.5 | 3123 | 1761.5 |
| 1096 | 2524 | 1462 | 2724 | 1562 | 2924 | 1662 | 3124 | 1762 |
| 1100 | 2525 | 1462.5 | 2725 | 1562.5 | 2925 | 1662.5 | 3125 | 1762.5 |
| 1104 | 2526 | 1463 | 2726 | 1563 | 2926 | 1663 | 3126 | 1763 |
| 1108 | 2527 | 1463.5 | 2727 | 1563.5 | 2927 | 1663.5 | 3127 | 1763.5 |
| 1112 | 2528 | 1464 | 2728 | 1564 | 2928 | 1664 | 3128 | 1764 |
| 1116 | 2529 | 1464.5 | 2729 | 1564.5 | 2929 | 1664.5 | 3129 | 1764.5 |
| 1120 | 2530 | 1465 | 2730 | 1565 | 2930 | 1665 | 3130 | 1765 |
| 1124 | 2531 | 1465.5 | 2731 | 1565.5 | 2931 | 1665.5 | 3131 | 1765.5 |
| 1128 | 2532 | 1466 | 2732 | 1566 | 2932 | 1666 | 3132 | 1766 |
| 1132 | 2533 | 1466.5 | 2733 | 1566.6 | 2933 | 1666.5 | 3133 | 1766.5 |
| 1136 | 2534 | 1467 | 2734 | 1567 | 2934 | 1667 | 3134 | 1767 |
| 1140 | 2535 | 1467.5 | 2735 | 1567.5 | 2935 | 1667.5 | 3135 | 1767.5 |
| 1144 | 2536 | 1468 | 2736 | 1568 | 2936 | 1668 | 3136 | 1768 |
| 1148 | 2537 | 1468.5 | 2737 | 1568.5 | 2937 | 1668.5 | 3137 | 1768.5 |
| 1152 | 2538 | 1469 | 2738 | 1569 | 2938 | 1669 | 3138 | 1769 |
| 1156 | 2539 | 1469.5 | 2739 | 1569.5 | 2939 | 1669.5 | 3139 | 1769.5 |
| 1160 | 2540 | 1470 | 2740 | 1570 | 2940 | 1670 | 3140 | 1770 |
| 1164 | 2541 | 1470.5 | 2741 | 1570.5 | 2941 | 1670.5 | 3141 | 1770.5 |
| 1168 | 2542 | 1471 | 2742 | 1571 | 2942 | 1671 | 3142 | 1771 |
| 1172 | 2543 | 1471.5 | 2743 | 1571.5 | 2943 | 1671.5 | 3143 | 1771.5 |
| 1176 | 2544 | 1472 | 2744 | 1572 | 2944 | 1672 | 3144 | 1772 |
| 1180 | 2545 | 1472.5 | 2745 | 1572.5 | 2945 | 1672.5 | 3145 | 1772.5 |
| 1184 | 2546 | 1473 | 2746 | 1573 | 2946 | 1673 | 3146 | 1773 |
| 1188 | 2547 | 1473.5 | 2747 | 1573.5 | 2947 | 1673.5 | 3147 | 1773.5 |
| 1192 | 2548 | 1474 | 2748 | 1574 | 2948 | 1674 | 3148 | 1774 |
| 1196 | 2549 | 1474.5 | 2749 | 1574.5 | 2949 | 1674.5 | 3149 | 1774.5 |
| 1200 | 2550 | 1475 | 2750 | 1575 | 2950 | 1675 | 3150 | 1775 |

**TABLE 30.1 (cont'd)**



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T03:58Z
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| Elapsed Hrs. of Operating Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chanl. | Freq. MHz | T2/R1 Chanl. | Freq. MHz | T3/R4 Chanl. | Freq. MHz | T4/R3 Chanl. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 12Ø4 | 2551 | 1475.5 | 2751 | 1575.5 | 2951 | 1675.5 | 3151 | 1775.5 |
| 12Ø8 | 2552 | 1476 | 2752 | 1576 | 2952 | 1676 | 3152 | 1776 |
| 1212 | 2553 | 1476.5 | 2753 | 1576.5 | 2953 | 1676.5 | 3153 | 1776.5 |
| 1216 | 2554 | 1477 | 2754 | 1577 | 2954 | 1677 | 3154 | 1777 |
| 122Ø | 2555 | 1477.5 | 2755 | 1577.5 | 2955 | 1677.5 | 3155 | 1777.5 |
| 1224 | 2556 | 1478 | 2756 | 1578 | 2956 | 1678 | 3156 | 1778 |
| 1228 | 2557 | 1478.5 | 2757 | 1578.5 | 2957 | 1678.5 | 3157 | 1778.5 |
| 1232 | 2558 | 1479 | 2758 | 1579 | 2958 | 1679 | 3158 | 1779 |
| 1236 | 2559 | 1479.5 | 2759 | 1579.5 | 2959 | 1679.5 | 3159 | 1779.5 |
| 124Ø | 256Ø | 148Ø | 276Ø | 158Ø | 296Ø | 168Ø | 316Ø | 178Ø |
| 1244 | 2561 | 148Ø.5 | 2761 | 158Ø.5 | 2961 | 168Ø.5 | 3161 | 178Ø.5 |
| 1248 | 2562 | 1481 | 2762 | 1581 | 2962 | 1681 | 3162 | 1781 |
| 1252 | 2563 | 1481.5 | 2763 | 1581.5 | 2963 | 1681.5 | 3163 | 1781.5 |
| 1256 | 2564 | 1482 | 2764 | 1582 | 2964 | 1682 | 3164 | 1782 |
| 126Ø | 2565 | 1482.5 | 2765 | 1582.5 | 2965 | 1682.5 | 3165 | 1782.5 |
| 1264 | 2566 | 1483 | 2766 | 1583 | 2966 | 1683 | 3166 | 1783 |
| 1268 | 2567 | 1483.5 | 2767 | 1583.5 | 2967 | 1683.5 | 3167 | 1783.5 |
| 1272 | 2568 | 1484 | 2768 | 1584 | 2968 | 1684 | 3168 | 1784 |
| 1276 | 2569 | 1484.5 | 2769 | 1584.5 | 2969 | 1684.5 | 3169 | 1784.5 |
| 128Ø | 257Ø | 1485 | 277Ø | 1585 | 297Ø | 1685 | 317Ø | 1785 |
| 1284 | 2571 | 1485.5 | 2771 | 1585.5 | 2971 | 1685.5 | 3171 | 1785.5 |
| 1288 | 2572 | 1486 | 2772 | 1586 | 2972 | 1686 | 3172 | 1786 |
| 1292 | 2573 | 1486.5 | 2773 | 1586.5 | 2973 | 1686.5 | 3173 | 1786.5 |
| 1296 | 2574 | 1487 | 2774 | 1587 | 2974 | 1687 | 3174 | 1787 |
| 13ØØ | 2575 | 1487.5 | 2775 | 1587.5 | 2975 | 1687.5 | 3175 | 1787.5 |
| 13Ø4 | 2576 | 1488 | 2776 | 1588 | 2976 | 1688 | 3176 | 1788 |
| 13Ø8 | 2577 | 1488.5 | 2777 | 1588.5 | 2977 | 1688.5 | 3177 | 1788.5 |
| 1312 | 2578 | 1489 | 2778 | 1589 | 2978 | 1689 | 3178 | 1789 |
| 1316 | 2579 | 1489.5 | 2779 | 1589.5 | 2979 | 1689.5 | 3179 | 1789.5 |
| 1320 | 258Ø | 149Ø | 278Ø | 159Ø | 298Ø | 169Ø | 318Ø | 179Ø |

TABLE 30.1 (cont'd)



Source: https://assist.dla.mil -- Downloaded: 2026Ø10-26T03:58Z
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| Elapsed Hrs. of Operating Time | RF TUNING HEAD CHANGE PAIRS T1/R2 Chan. | Freq. MHz | T2/R1 Chan. | Freq. MHz | T3/R4 Chan. | Freq. MHz | T4/R3 Chan. | Freq. MHz |
|---|---|---|---|---|---|---|---|---|
| 1324 | 2581 | 1490.5 | 2781 | 1590.5 | 2981 | 1690.5 | 3181 | 1790.5 |
| 1328 | 2582 | 1491 | 2782 | 1591 | 2982 | 1691 | 3182 | 1791 |
| 1332 | 2583 | 1491.5 | 2783 | 1591.5 | 2983 | 1691.5 | 3183 | 1791.5 |
| 1336 | 2584 | 1492 | 2784 | 1592 | 2984 | 1692 | 3184 | 1792 |
| 1340 | 2585 | 1492.5 | 2785 | 1592.5 | 2985 | 1692.5 | 3185 | 1792.5 |
| 1344 | 2586 | 1493 | 2786 | 1593 | 2986 | 1693 | 3186 | 1793 |
| 1348 | 2587 | 1493.5 | 2787 | 1593.5 | 2987 | 1693.5 | 3187 | 1793.5 |
| 1352 | 2588 | 1494 | 2788 | 1594 | 2988 | 1694 | 3188 | 1794 |
| 1356 | 2589 | 1494.5 | 2789 | 1594.5 | 2989 | 1694.5 | 3189 | 1794.5 |
| 1360 | 2590 | 1495 | 2790 | 1595 | 2990 | 1695 | 3190 | 1795 |
| 1364 | 2591 | 1495.5 | 2791 | 1595.5 | 2991 | 1695.5 | 3191 | 1795.5 |
| 1368 | 2592 | 1496 | 2792 | 1596 | 2992 | 1696 | 3192 | 1796 |
| 1372 | 2593 | 1496.5 | 2793 | 1596.5 | 2993 | 1696.5 | 3193 | 1796.5 |
| 1376 | 2594 | 1497 | 2794 | 1597 | 2994 | 1697 | 3194 | 1797 |
| 1380 | 2595 | 1497.5 | 2795 | 1597.5 | 2995 | 1697.5 | 3195 | 1797.5 |
| 1384 | 2596 | 1498 | 2796 | 1598 | 2996 | 1698 | 3196 | 1798 |
| 1388 | 2597 | 1498.5 | 2797 | 1598.5 | 2997 | 1698.5 | 3197 | 1798.5 |
| 1392 | 2598 | 1499 | 2798 | 1599 | 2998 | 1699 | 3198 | 1799 |
| 1396 | 2599 | 1499.5 | 2799 | 1599.5 | 2999 | 1699.5 | 3199 | 1799.5 |
| 1400 | 2600 | 1500 | 2800 | 1600 | 3000 | 1700 | 3200 | 1800 |
| 1404 | 2601 | 1500.5 | 2801 | 1600.5 | 3001 | 1700.5 | 3201 | 1800.5 |
| 1408 | 2602 | 1501 | 2802 | 1601 | 3002 | 1701 | 3202 | 1801 |
| 1412 | 2603 | 1501.5 | 2803 | 1601.5 | 3003 | 1701.5 | 3203 | 1801.5 |
| 1416 | 2604 | 1502 | 2804 | 1602 | 3004 | 1702 | 3204 | 1802 |
| 1420 | 2605 | 1502.5 | 2805 | 1602.5 | 3005 | 1702.5 | 3205 | 1802.5 |
| 1424 | 2606 | 1503 | 2806 | 1603 | 3006 | 1703 | 3206 | 1803 |
| 1428 | 2607 | 1503.5 | 2807 | 1603.5 | 3007 | 1703.5 | 3207 | 1803.5 |
| 1432 | 2608 | 1504 | 2808 | 1604 | 3008 | 1704 | 3208 | 1804 |
| 1436 | 2609 | 1504.5 | 2809 | 1604.5 | 3009 | 1704.5 | 3209 | 1804.5 |
| 1440 | 2610 | 1505 | 2810 | 1605 | 3010 | 1705 | 3210 | 1805 |

**TABLE 30.1 (cont'd)**



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FIGURE 30.1 RX FIXED HEAD

| DATE | TIME | RF HEAD NO. | CH. NO. | RECEIVER METER READINGS (F.S.D.) | | | | | | | | | | | O.W. CHECK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 180 KHz | 300 KHz | OW | +12V DC | -12V DC | REFL PWR | XMTR OUTL | OSC | DOUB | MULT | RCVR DIS | |
| LIMITS | | | | 40-60% | 40-60% | 40-60% | 40-60% | 40-60% | less than 15% | above 40% | 20-90% | 20-90% | above 25% | 30-90% | Light on Buzzer on (1600 Hz) |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
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FIGURE 30.1 (cont'd) TX FIXED HEAD

| DATE | TIME | RF HEAD No. | CH. No. | TRANSMITTER METER READINGS (F.S.D.) | | | | | | | | | | | O.W. CHECK |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 180 KHz 12CH | 300 KHz 24CH | 12V DC | 28V DC | 600V DC | OSC | DOUB | MULT | DRIV | PWR OUT | REFL PWR | |
| LIMITS | | | | 40-60% | 40-60% | 40-60% | 40-60% | 40-60% | 20-90% | 20-90% | 20-90% | 20-90% | above 40% | less than 15% | Light on Buffer on (1600 Hz) |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
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FIGURE 30.2 FAILURE DATA SHEET TECHNICIAN _____

PART 1 - TO BE COMPLETED AT TIME OF FAILURE

1. F.H./R.F. HEAD COMBINATION _____/_____

2. DAYS/HOURS FROM START _____/_____

3. CHANNEL AT TIME OF FAILURE Ch. _____

4. SYMPTOMS OF FAILURE:

_____

_____

_____

_____

5. UNIT REMOVED FROM TEST (F.H. OR R.F. HEAD) _____

6. F.R.F. No. _____

7. TOTAL DOWN-TIME OF FAILED UNIT _____

PART 2 - TO BE COMPLETED WHEN F.R.F. CLOSED

1. SUB-UNIT FAILED _____

2. DWG. No. _____

3. SERIAL No. _____

4. COMPONENT REPLACED _____

5. COMMENTS:

_____

_____

_____

Signature _____

Date _____



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## SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION NUMBER: MIL-R-[illegible](EL) RADIO SET AN/GRC-[illegible]

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 [illegible]

ESC-FM 1068-68

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U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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U.S.Army Electronics Command
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